

PAULEE GROUP · CENTER FOR QUANTUM TECHNOLOGY, KIST

Distributed Quantum Machine Learning

Three quantum processors, classical communication, and what I doubt

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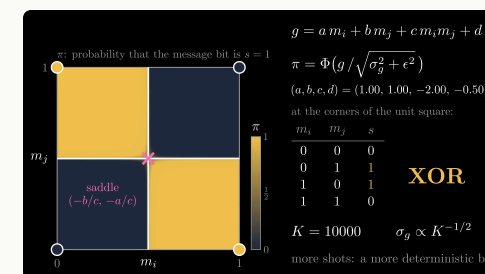
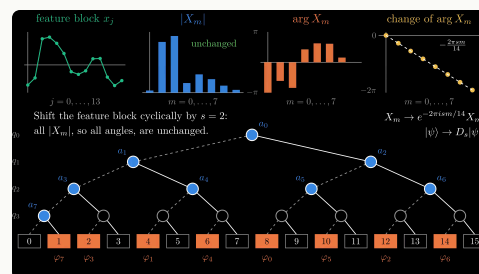
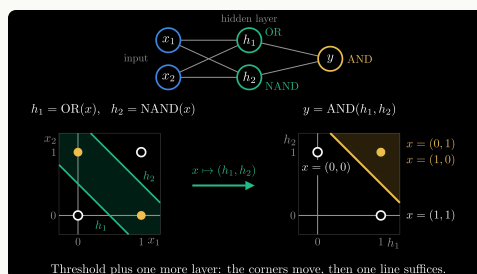
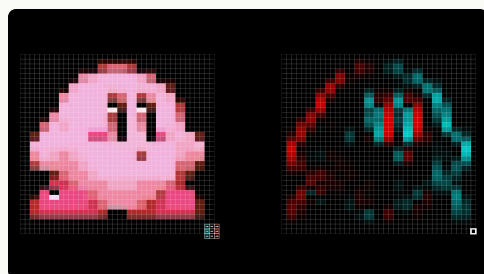
2026-09-29 · Lab Meeting

Shout-out to Manim



The animations in this talk were made with **Manim**.

- **ManimGL 1.7.2** by Grant Sanderson (3Blue1Brown): his own convolution scenes, on Kirby
- **Manim Community Edition v0.21.0**: all other clips



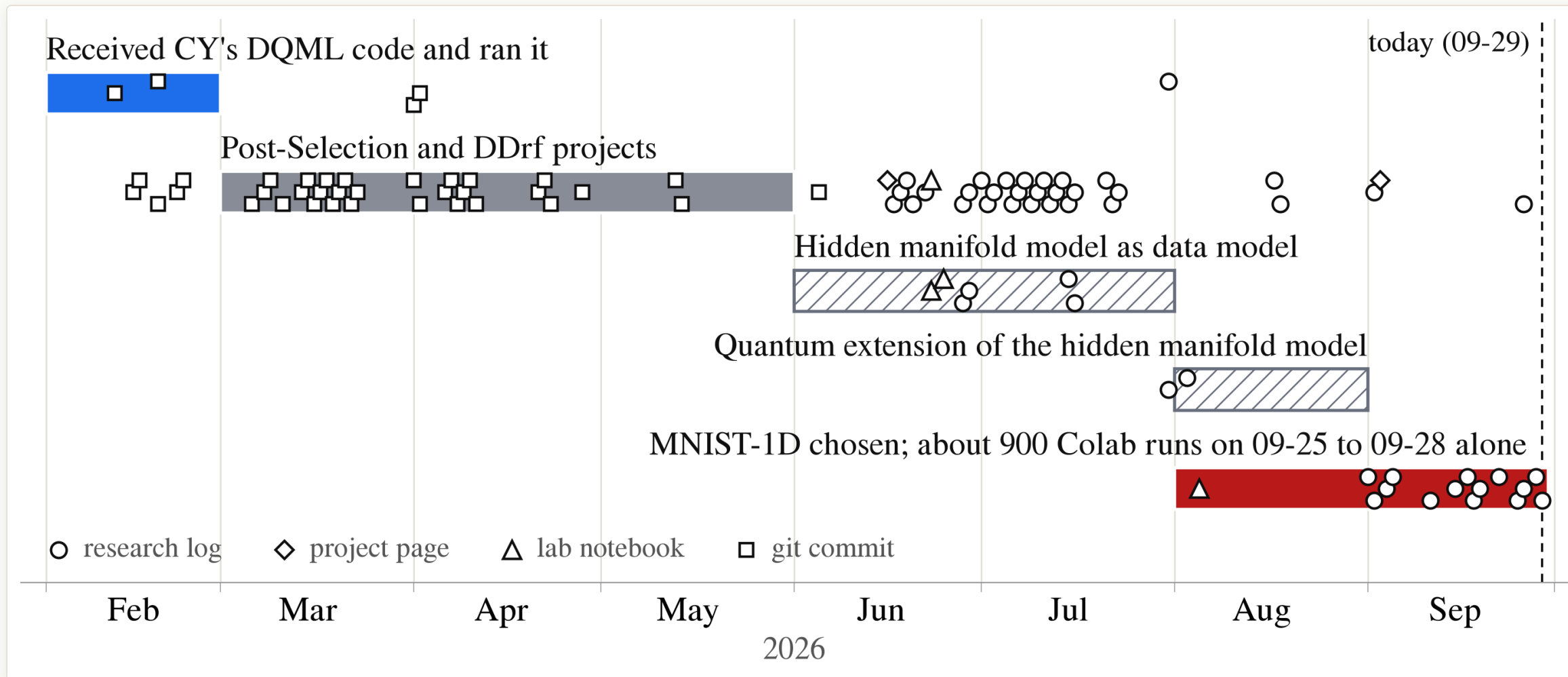
Convolution scenes: github.com/3b1b/videos (CC BY-NC-SA 4.0). Manim: github.com/3b1b/manim, github.com/ManimCommunity/manim (MIT).

Outline

- 0. Where the months went
- 1. Introduction to machine learning
- 2. Quantum and distributed machine learning
- 3. Data: MNIST-1D
- 4. The model
- 5. Results
- 6. Skepticism and a suggestion

INTRO METHOD STRATEGY RESULTS ONGOING

Where the months went



01

Introduction to machine learning

Learning, nonlinearity, convolution

What is machine learning?

Model. A function f_θ with trainable parameters θ .

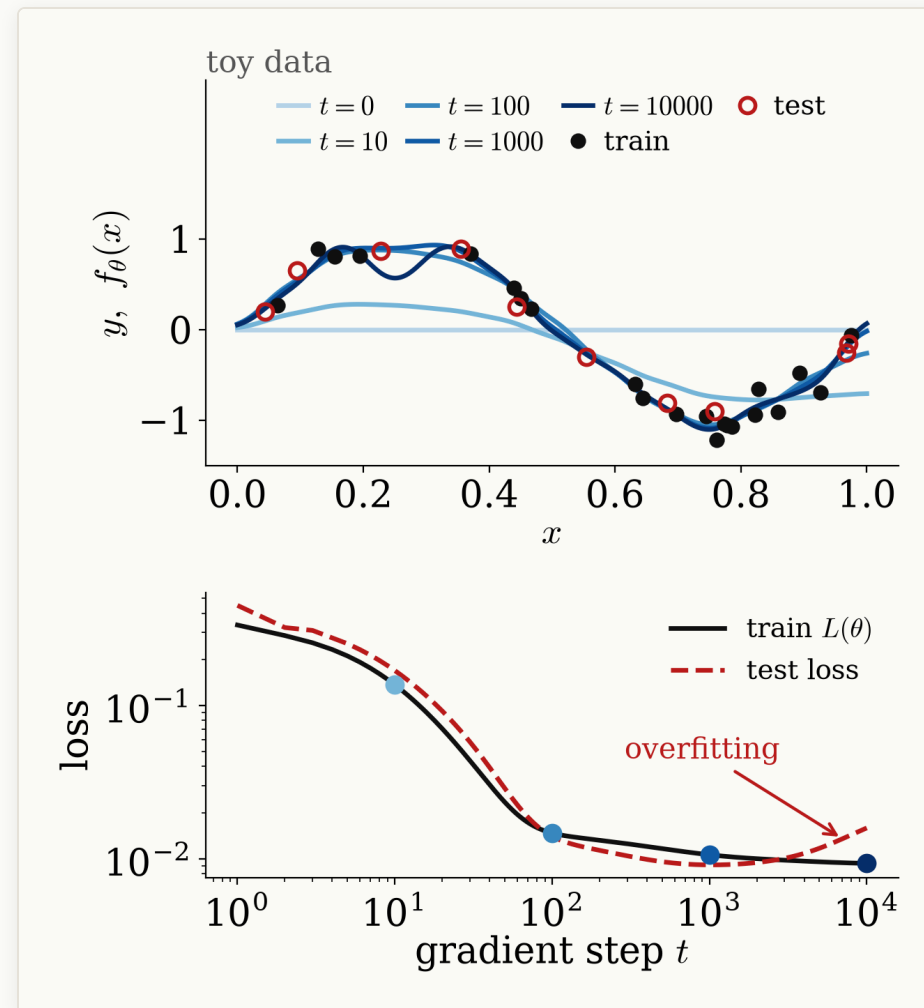
Loss on M training examples (x_i, y_i) :

$$L(\theta) = \frac{1}{M} \sum_{i=1}^M (f_\theta(x_i) - y_i)^2$$

Training. Gradient descent with step size η :

$$\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$$

Success is judged on unseen (test) data.



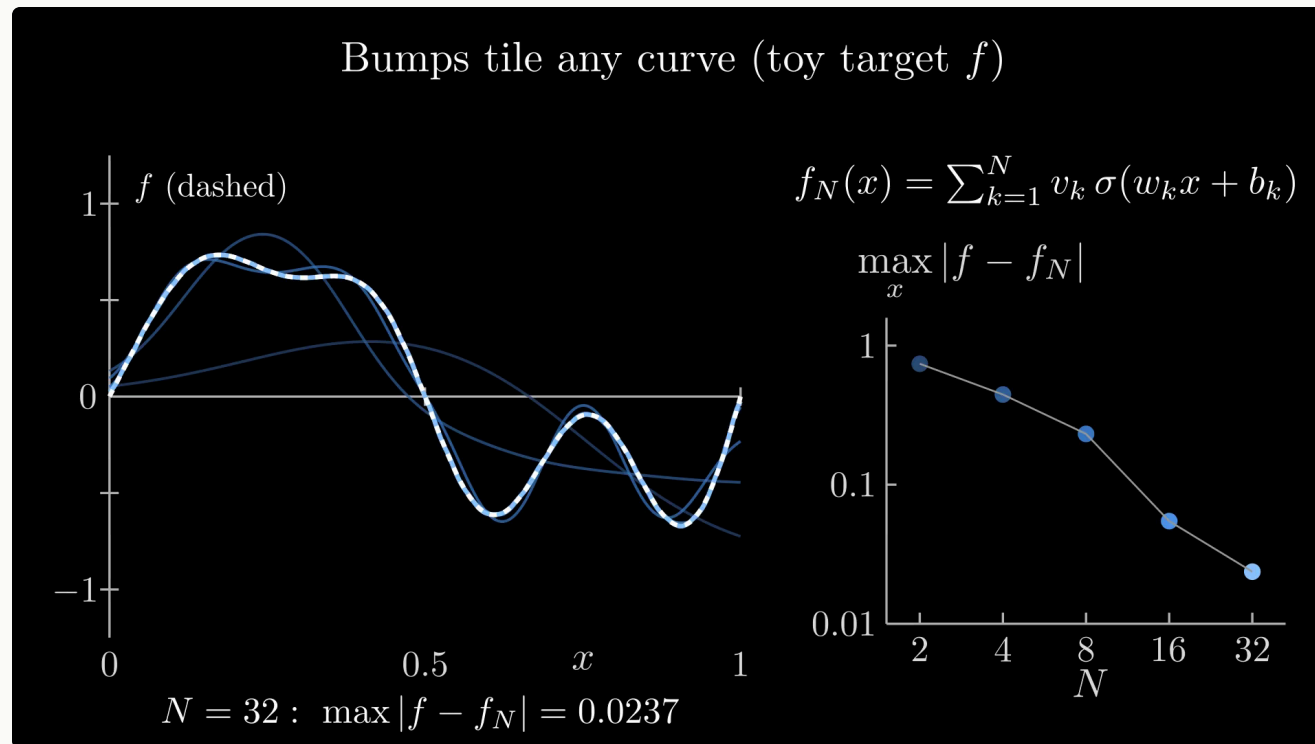
Universal approximation theorem

For continuous f on a compact $K \subset \mathbb{R}^d$ and any $\varepsilon > 0$, some N and v_k, w_k, b_k give

$$\sup_{x \in K} \left| f(x) - \sum_{k=1}^N v_k \sigma(w_k \cdot x + b_k) \right| < \varepsilon.$$

- Two steep sigmoids make a bump; weighted bumps tile any curve.
- The theorem does not say how to find the weights or how large N must be.

Cybenko, Math. Control Signals Syst. 2, 303 (1989); Hornik, Neural Netw. 4, 251 (1991).

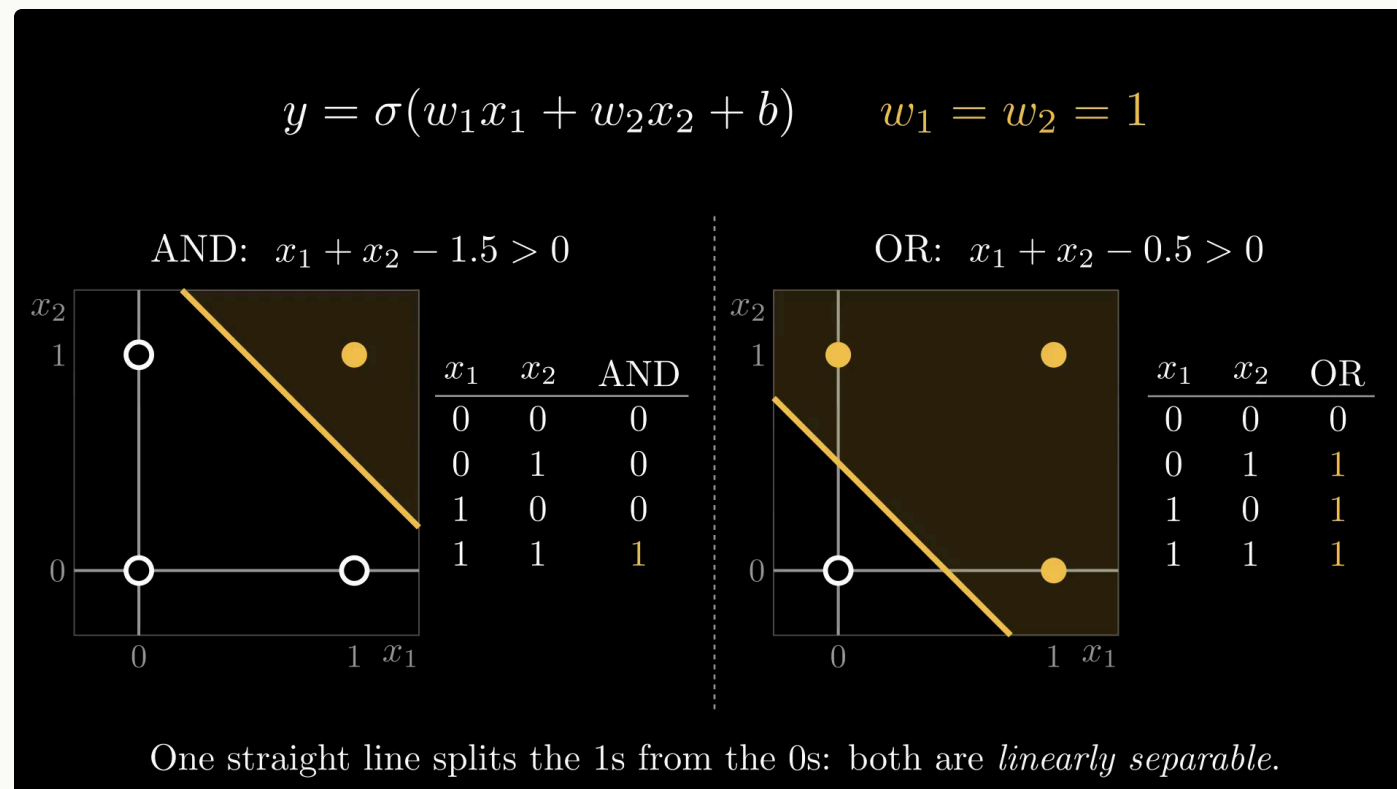


One neuron: a weighted sum and a threshold

$$y = \sigma(w_1x_1 + w_2x_2 + b), \quad \sigma(z) = \begin{cases} 1, & z > 0 \\ 0, & z \leq 0 \end{cases}$$

- The **decision boundary**
 $w_1x_1 + w_2x_2 + b = 0$ is a line.
- $w = (1, 1)$: $b = -1.5$ gives AND,
 $b = -0.5$ gives OR.
- Both are **linearly separable**.

McCulloch and Pitts, Bull. Math. Biophys. 5, 115 (1943); Rosenblatt, Psychol. Rev. 65, 386 (1958).



XOR needs a threshold and one more layer

No single line separates XOR: its 0 corners need $w_1 + w_2 + 2b \leq 0$, its 1 corners $w_1 + w_2 + 2b > 0$.

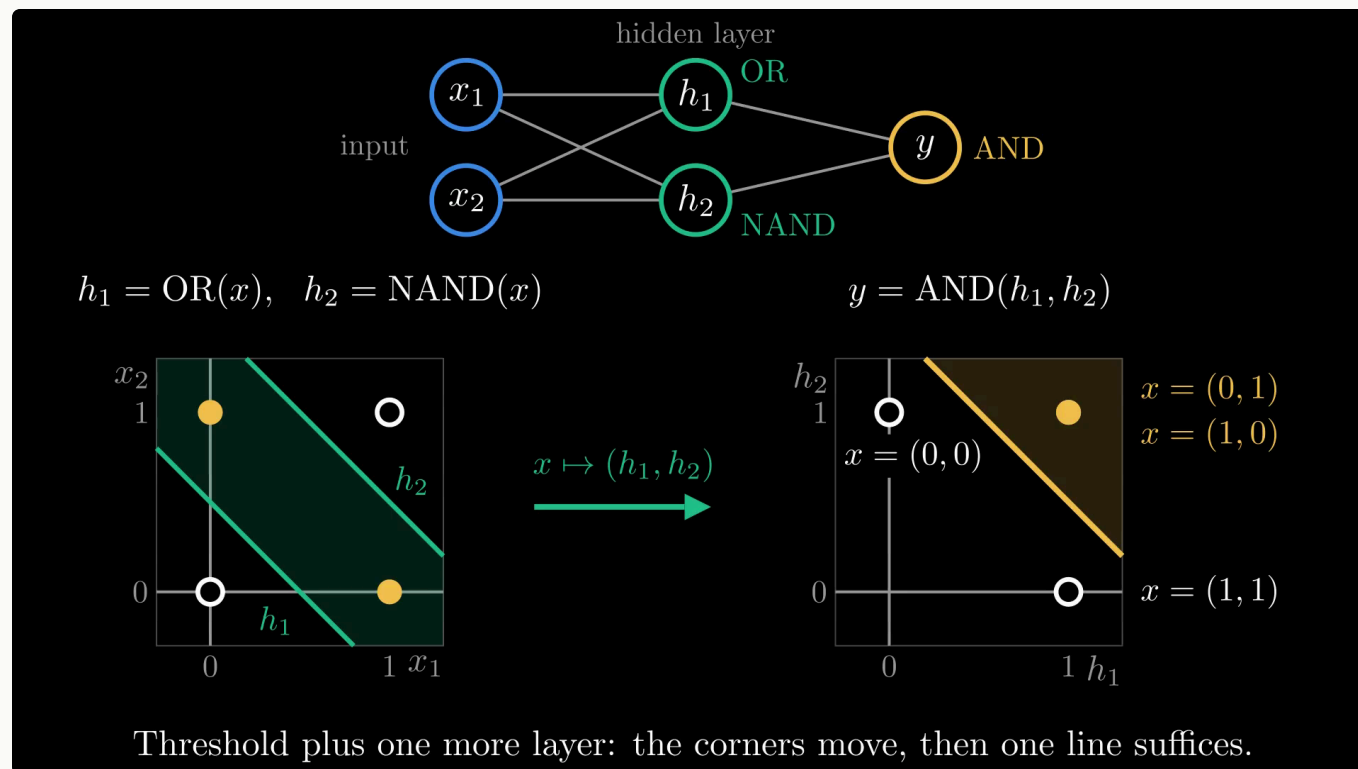
One hidden layer fixes it:

$$h_1 = \text{OR}(x), \quad h_2 = \text{NAND}(x)$$

$$y = \text{AND}(h_1, h_2)$$

Without the threshold, two linear layers collapse into one.

Minsky and Papert, *Perceptrons*, MIT Press (1969); Rumelhart, Hinton and Williams, *Parallel Distributed Processing*, Vol. 1, MIT Press (1986).



A fully connected network: layers of neurons

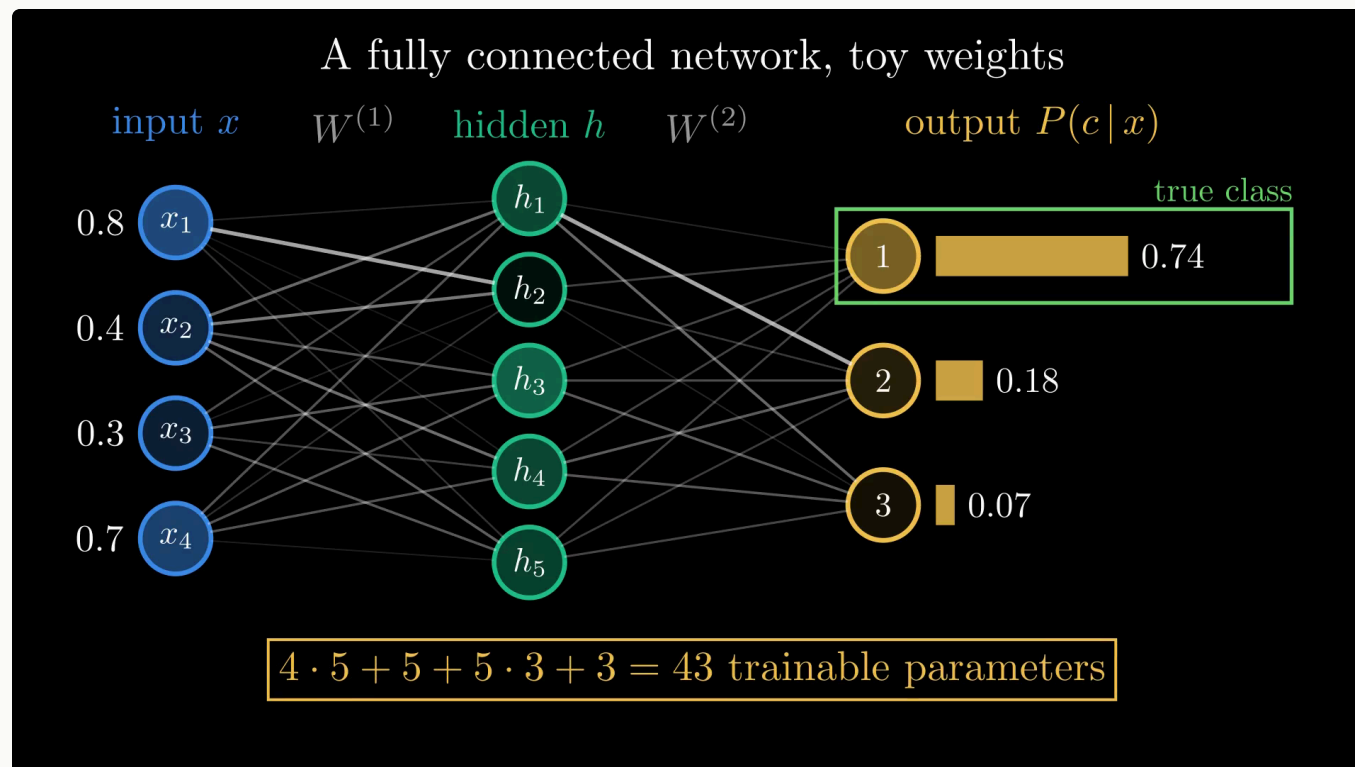
$$h = \sigma(W^{(1)}x + b^{(1)}), \quad \sigma(z) = 1/(1 + e^{-z})$$

$$P(c | x) = \text{softmax}_c(W^{(2)}h + b^{(2)})$$

$$L = -\frac{1}{M} \sum_i \log P(y_i | x_i)$$

- **Fully connected:** all W_{ij} are free.
- **Training:** gradient descent on the cross-entropy L ; backpropagation gives $\nabla_{\theta} L$.
- **Size:** the MNIST-1D reference MLP has 15 210 parameters.

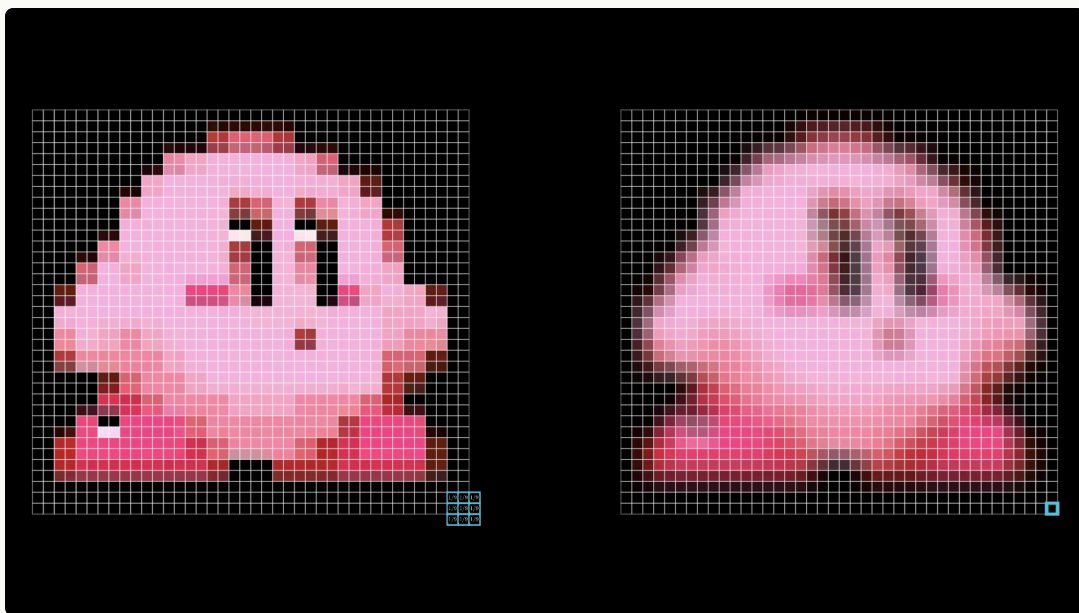
Rumelhart, Hinton and Williams, Nature 323, 533 (1986); Goodfellow, Bengio and Courville, *Deep Learning* (2016), ch. 6.



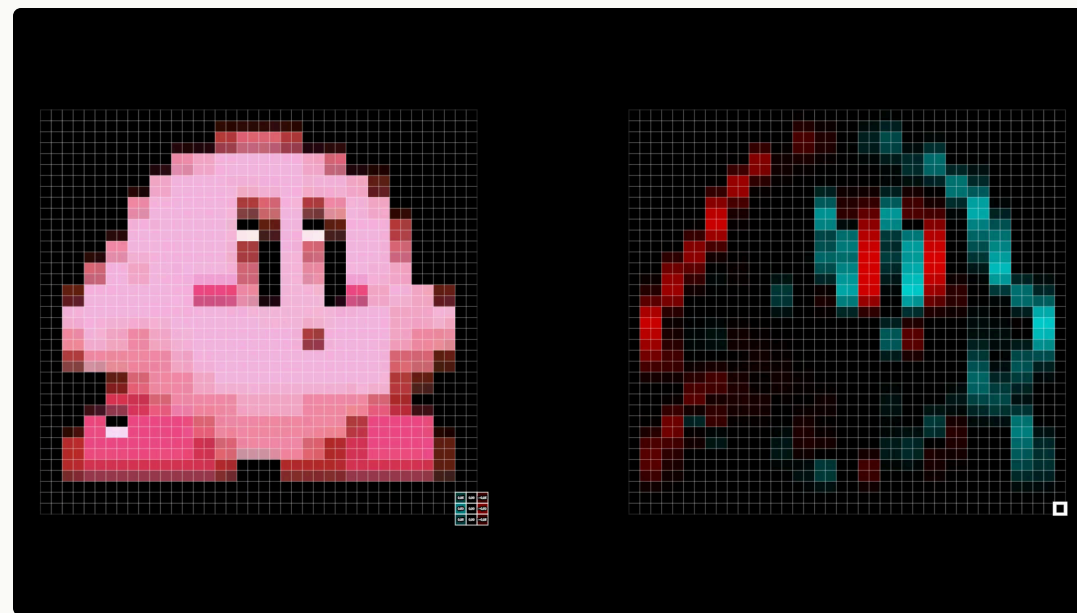
Convolution extracts local features

$(x * k)[i, j] = \sum_{m, n} k_{m, n} x_{i+m, j+n}$ Each output pixel weights the patch under the kernel k ; the same k at every (i, j) gives translation equivariance.

Box blur, $k_{m, n} = 1/9$



Sobel kernel: vertical edges



A CNN learns its kernels. MNIST-1D, ten digits (published): CNN ≈ 0.94 , MLP ≈ 0.68 .

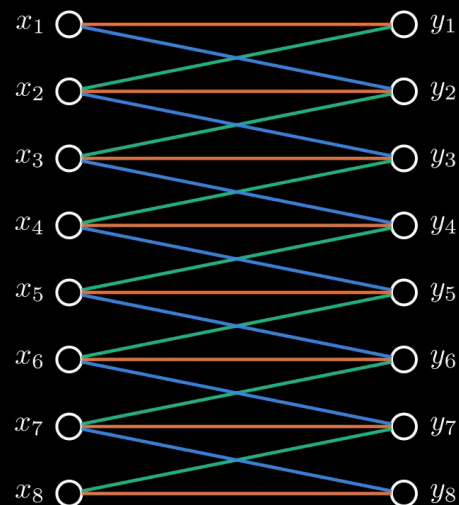
Clips: 3Blue1Brown, *But what is a convolution?* (2022), re-rendered with Kirby; see also *But what is a neural network?* (2017).

A CNN is a constrained fully connected network

A convolution is a fully connected (FC) layer $y = Wx$ with two constraints:

- **Locality:** $W_{ij} = 0$ outside the kernel, so W is banded.
- **Weight sharing:** $W_{ij} = k_{j-i}$, so W is Toeplitz.
- On 40 inputs, 1600 free weights become 5 (a width-5 kernel).

A convolutional layer is a constrained fully connected layer
free weights: 3



toy kernel

■ $k_{-1} = -0.6$

■ $k_0 = 1.0$

■ $k_{+1} = 0.4$

0.4	-0.6	0	0	0	0	0	0
-0.6	0.4	0	0	0	0	0	0
0	-0.6	0.4	0	0	0	0	0
0	0	-0.6	0.4	0	0	0	0
0	0	0	-0.6	0.4	0	0	0
0	0	0	0	-0.6	0.4	0	0
0	0	0	0	0	-0.6	0.4	0
0	0	0	0	0	0	-0.6	0.4

row i = output y_i , column j = input x_j

conv = FC with most weights zero and the rest tied

Three objectives

1. **Distributed vs monolithic:** three 4-qubit QPUs against one 12-qubit QPU.
2. **Classical communication:** what do the bits sent between QPUs add, and what do they compute?
3. **Topology:** does it matter which QPU sends to which?

Setting: Hwang et al., Quantum Sci. Technol. 10, 015059 (2025).

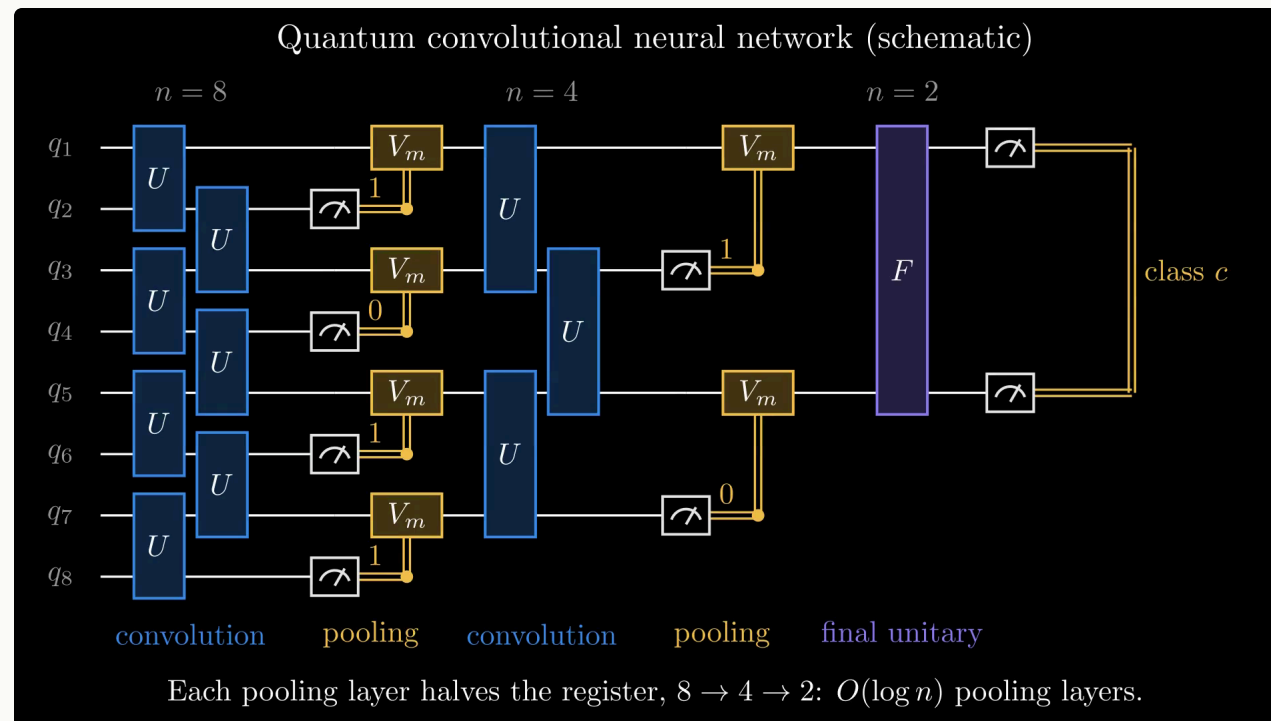
02

Quantum and distributed machine learning

From one quantum processor to three

The quantum convolutional neural network (QCNN)

- **Convolution:** quasi-local parametrised unitaries U
- **Pooling:** measure qubits; the outcome m controls a rotation V_m on a neighbour
- **Output:** a final unitary F , then measure the class



Nonlinearity from measurement, depth from pooling

Non-selective measurement: linear

$$P(c) = \text{Tr} [\Pi_c \mathcal{E}(\rho(x))] = \text{Tr} [E_c \rho(x)]$$

- Gates, measurements and feed-forward, averaged over outcomes: one channel $\mathcal{E}$
- Nonlinear in x only through the embedding $x \mapsto \rho(x)$

Selective measurement: nonlinear

$$\rho_\mu = \frac{M_\mu \rho M_\mu^\dagger}{\text{Tr} [M_\mu \rho M_\mu^\dagger]}$$

- Nonlinear in ρ through the normalisation
- A threshold $\hat{m} > \tau$ of a K -shot estimate: nonlinear, needs many copies

Depth: each pooling layer builds new features from the previous outcomes.

Cong, Choi, Lukin, Nat. Phys. 15, 1273 (2019).

Skepticism: four reasons for doubt

1. **Classical baselines win overall** on 160 small datasets; entanglement often does not help.
2. **Barren plateaus**: random circuits have gradients exponentially small in the qubit number.
3. **Trainable often means simulable**: classical surrogates match QCNNs up to 1024 qubits.
4. **The fine print**: data loading and readout can erase the speed-up.

[1] Bowles et al., arXiv (2024). [2] McClean et al., Nat. Commun. (2018). [3] Cerezo et al., Nat. Commun. (2025); Bermejo et al., PRX Quantum (2026). [4] Aaronson, Nat. Phys. (2015).

Optimism: reasons to keep going

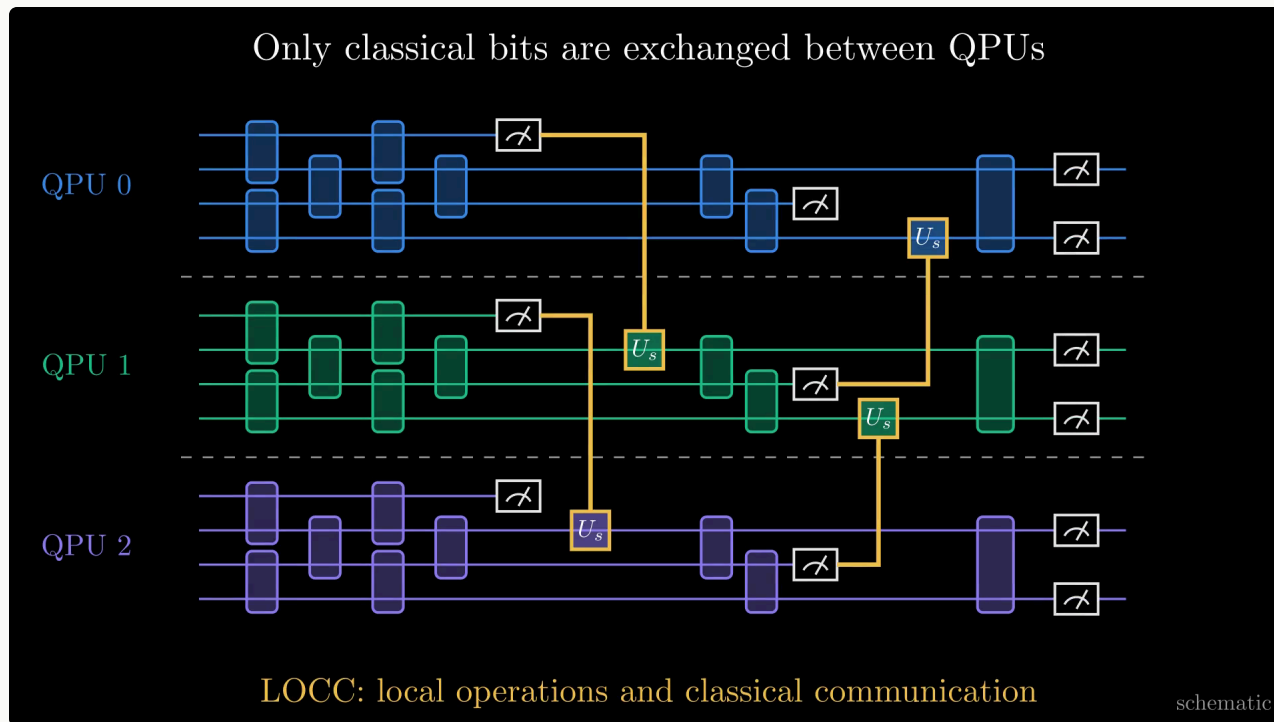
1. **Deep circuits can generalize:** double descent, also on MNIST-1D, while trainable.
2. **QCNNs have no barren plateaus:** the gradient variance decays only polynomially.
3. **Provable separations,** for special problems such as discrete logarithms.
4. **Spectral methods suit quantum computers:** the quantum Fourier transform is efficient.

Our aim: understand what communication does, not a quantum advantage.

[1] Kempkes et al. (2026). [2] Pesah et al., PRX (2021). [3] Liu et al., Nat. Phys. (2021); Huang et al., Science (2022). [4] Belis et al. (2026).

Distributed QML: one QCNN partitioned across several QPUs

- Entanglement between QPUs: proof-of-principle.
- Classical feed-forward: fast enough.
- A mid-circuit outcome on one QPU controls a gate on another.



Two 4-qubit QPUs, synthetic binary task: classical communication beats no communication by 5 to 11 points and matches quantum communication.

Hwang et al., Quantum Sci. Technol. 10, 015059 (2025); three-QPU extension: CY (previous DQML project). See also Chinzei et al., PRR 6, 023042 (2024).

Communication between QPUs: $\text{NC} \subset \text{CC} \subset \text{QC}$

communication	exchanged between QPUs	entanglement between QPUs
NC : no communication	nothing	none
CC : classical communication (LOCC), our model	message bits that control gates	none
QC : quantum communication	the pooled qubits	yes
monolithic QPU	(no partition)	yes

- **Deferred measurement**: a qubit used only as a control is worth one bit,

$$\text{Tr}_q[CU \rho CU^\dagger] = \sum_{\mu=0,1} U^\mu \langle \mu | \rho | \mu \rangle_q U^{\mu\dagger}.$$
- **Holevo bound**: without shared entanglement, one qubit carries at most one bit.

QC helps only if the receiving QPU processes the received qubits coherently.

Nielsen and Chuang (2010), §4.4, §12.1. QC in Hwang et al. (2025): non-local two-qubit gates.

How we test the objectives

1. **Distributed vs monolithic:** one 12-qubit QCNN with the same encoding.
2. **Classical communication:** accuracy against no communication (NC); which Boolean function each trained message bit computes.
3. **Topology:** accuracy per transmitted bit for $1 \rightarrow 1$, $2 \rightarrow 1$ (default), $2 \rightarrow 2$ (to senders), $2 \rightarrow 3$ (broadcast).
4. **Open, bits vs qubits:** send the pooled qubits (QC) or only their outcomes (CC), at the same cost and location.
5. **Open, lottery tickets:** remove random message edges, retrain: can a sparser graph do better? (not run yet)

Frankle and Carbin, ICLR (2019).

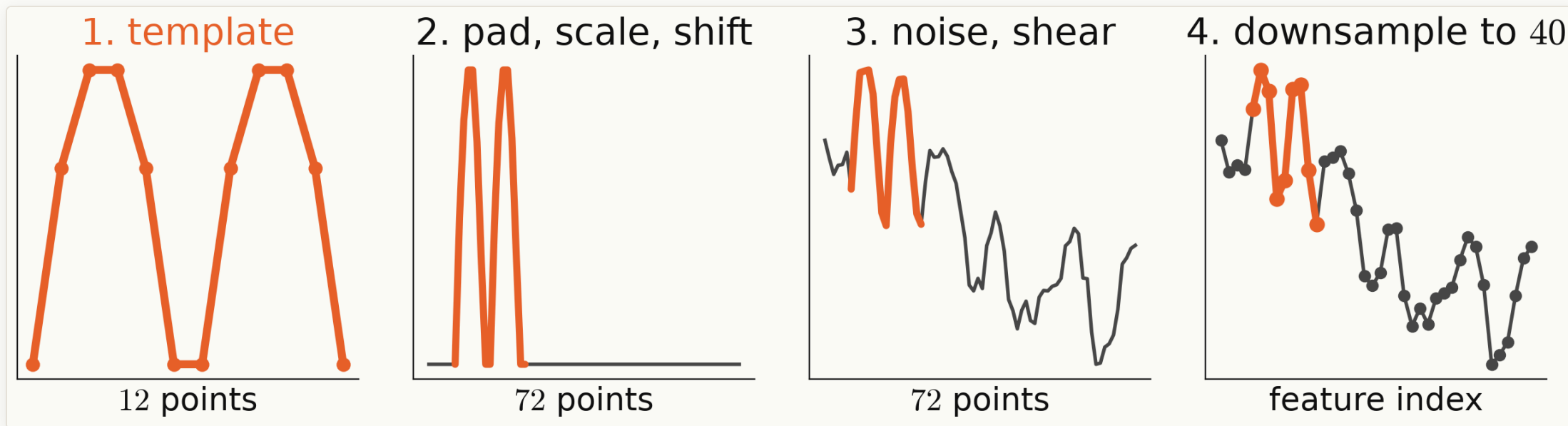
03

Data: MNIST-1D

A small benchmark, and classical baselines

METHOD MNIST-1D: a digit written as 40 numbers

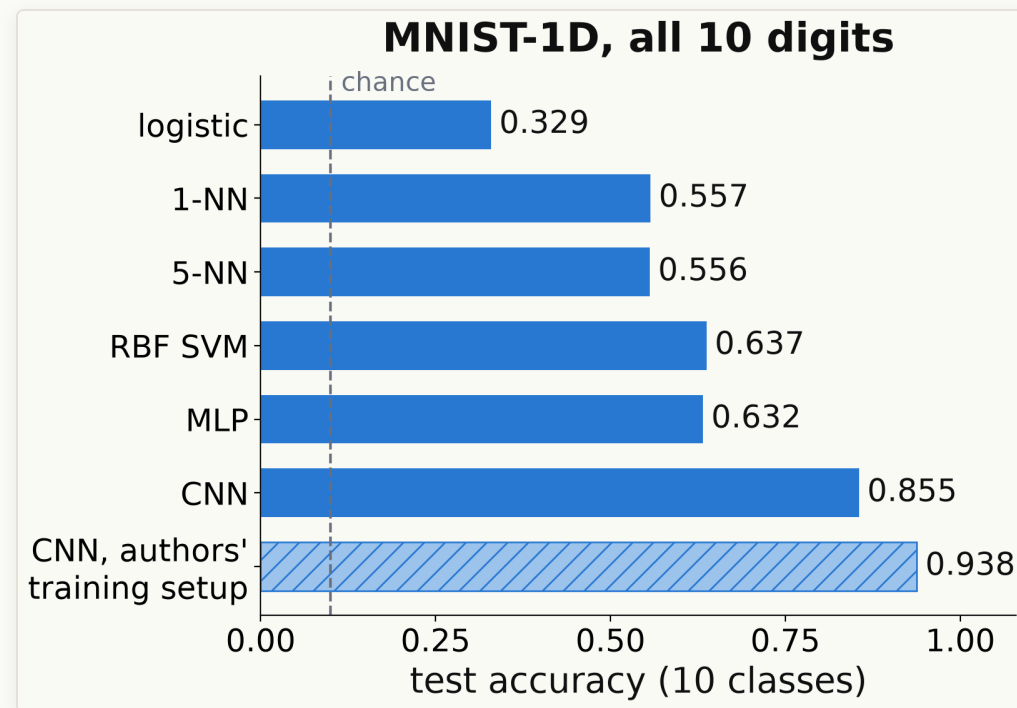
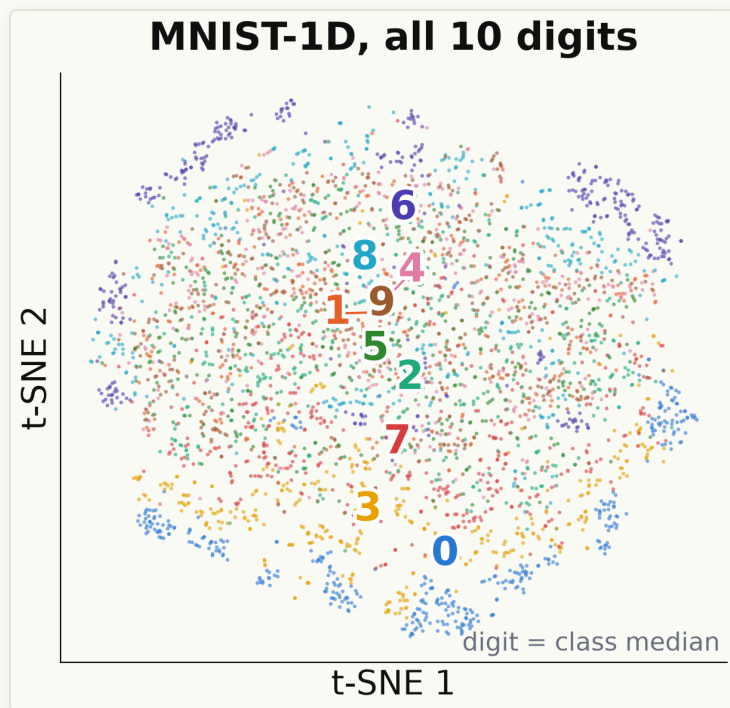
- Each signal: one of ten 12-point templates, randomly transformed to 40 features.
- The shift puts the digit anywhere: the label lives in the shape of a local stroke.
- Standard set: all 10 digits, 4000 training and 1000 test signals.



Greydanus and Kobak, ICML 2024 (arXiv:2011.14439).

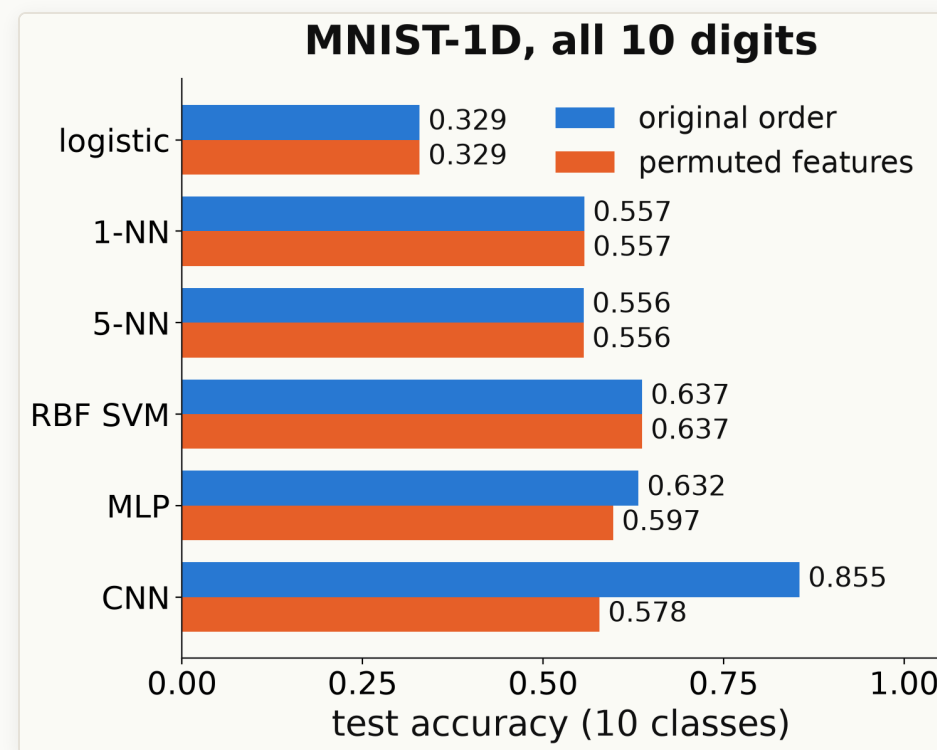
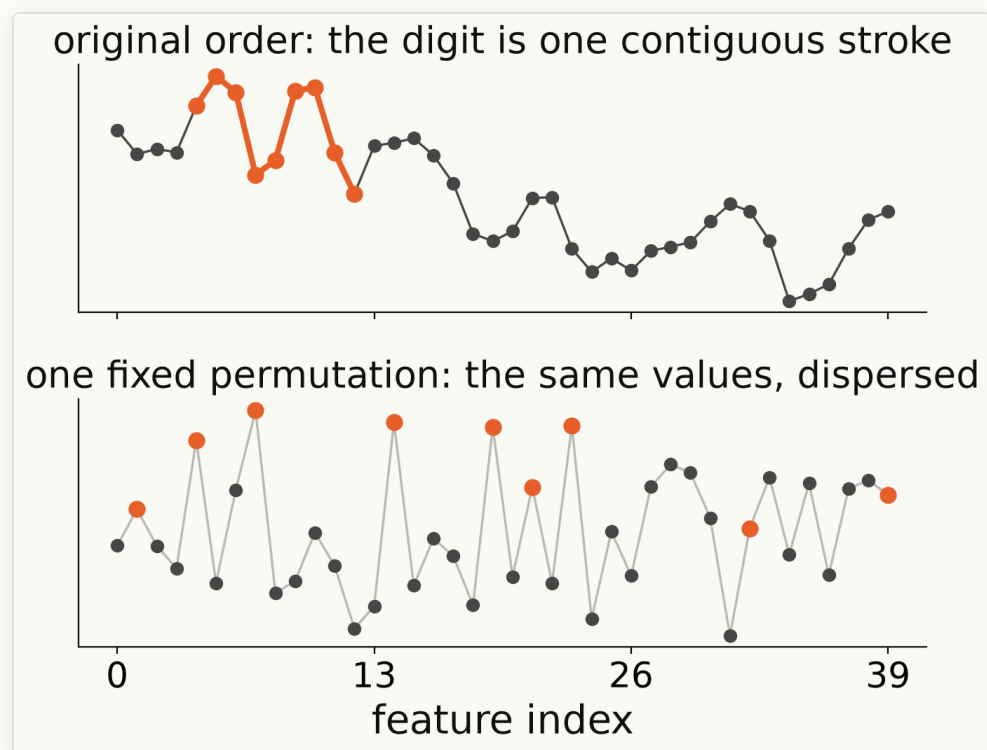
Why it is hard: no clusters, only local structure

- All 10 digits, raw features: no clusters (silhouette -0.039); two principal components hold 35.7% of the variance.
- Test accuracy, 10 digits: Gaussian-kernel (RBF) SVM 0.637, MLP 0.632, CNN 0.855 (0.938, authors' setup).
- The CNN wins by locality: small filters, reused at every position.

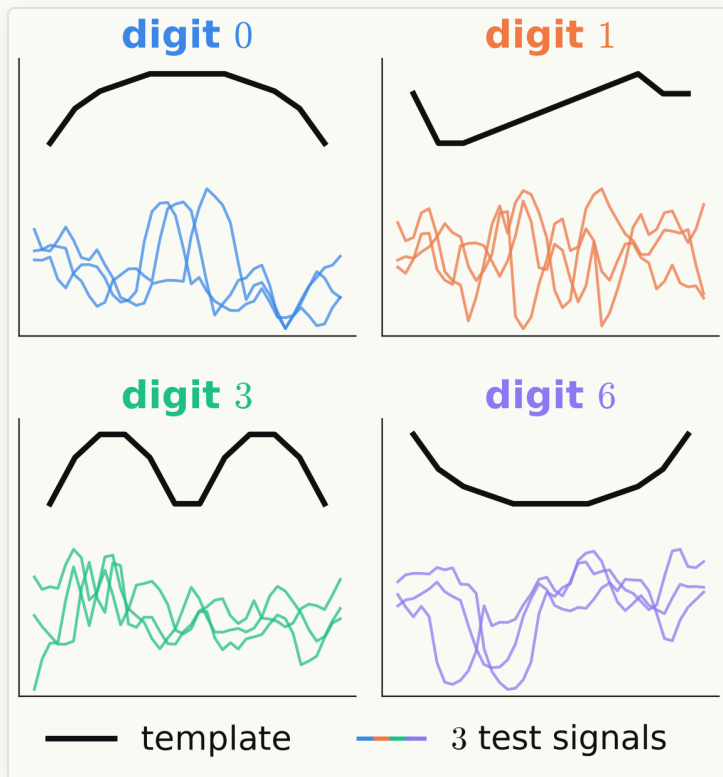


The feature-permutation test: the value of the locality prior

- One fixed random permutation of the 40 features ("shuffled MNIST-1D"); every model retrained.
- Logistic regression, k -nearest neighbours, RBF SVM unchanged: distances and inner products are preserved.
- On all 10 digits the CNN falls from 0.855 to 0.578: its **locality prior** is worth 0.277.



Our task: four digits, three feature blocks



- Digits 0, 1, 3, 6 only: 1325 training, 264 validation, 411 test signals.
- Each QPU encodes a block x_b of 14 features, contiguous or permuted.
- Additive model $\log P(c | x) = \sum_b f_b(c, x_b) - \log Z(x)$: the classical product of experts.

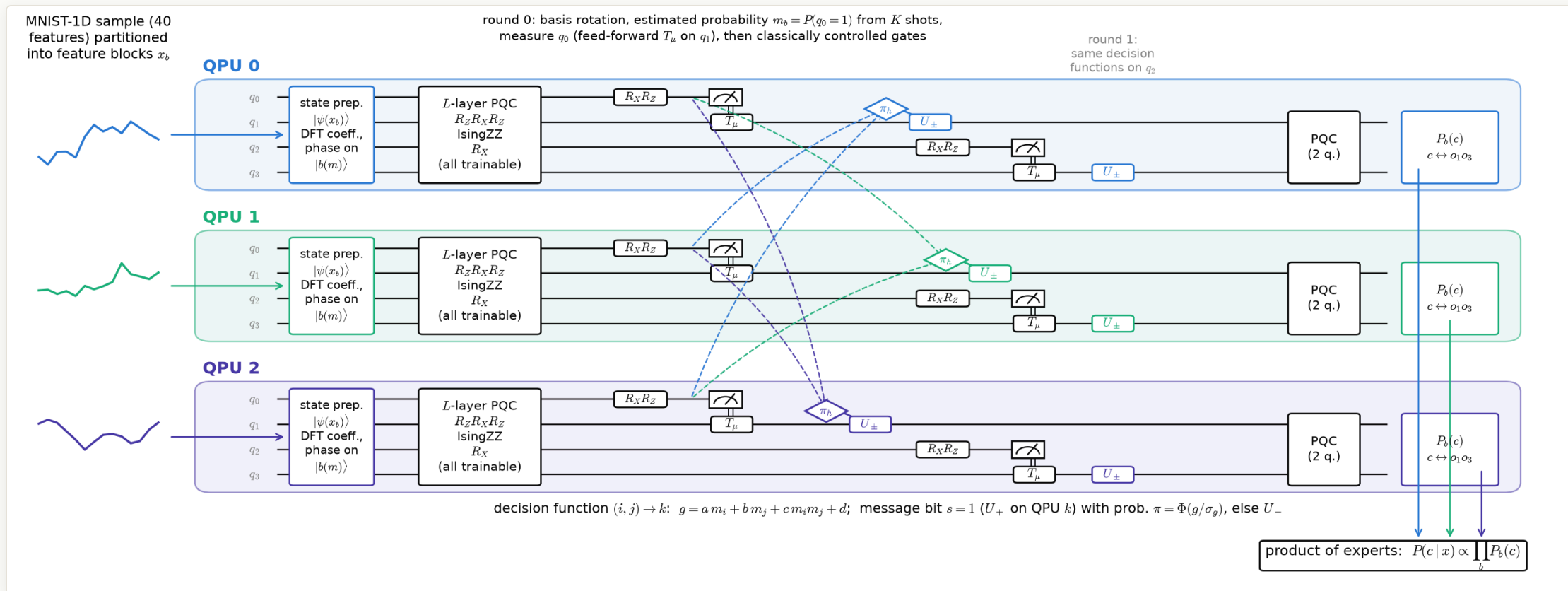
classical model, 4 digits	test accuracy
kernel logistic regression, all 40 features shift-aware kernel (sum of RBF kernels over cyclic width-5 sub-windows), chosen on validation data among 2330 models	0.978
tuned RBF SVM, all 40 features	0.964
additive, contiguous blocks	0.971
additive, permuted features	0.891

04

The model

Three four-qubit QPUs, classical messages only

The whole model on one page

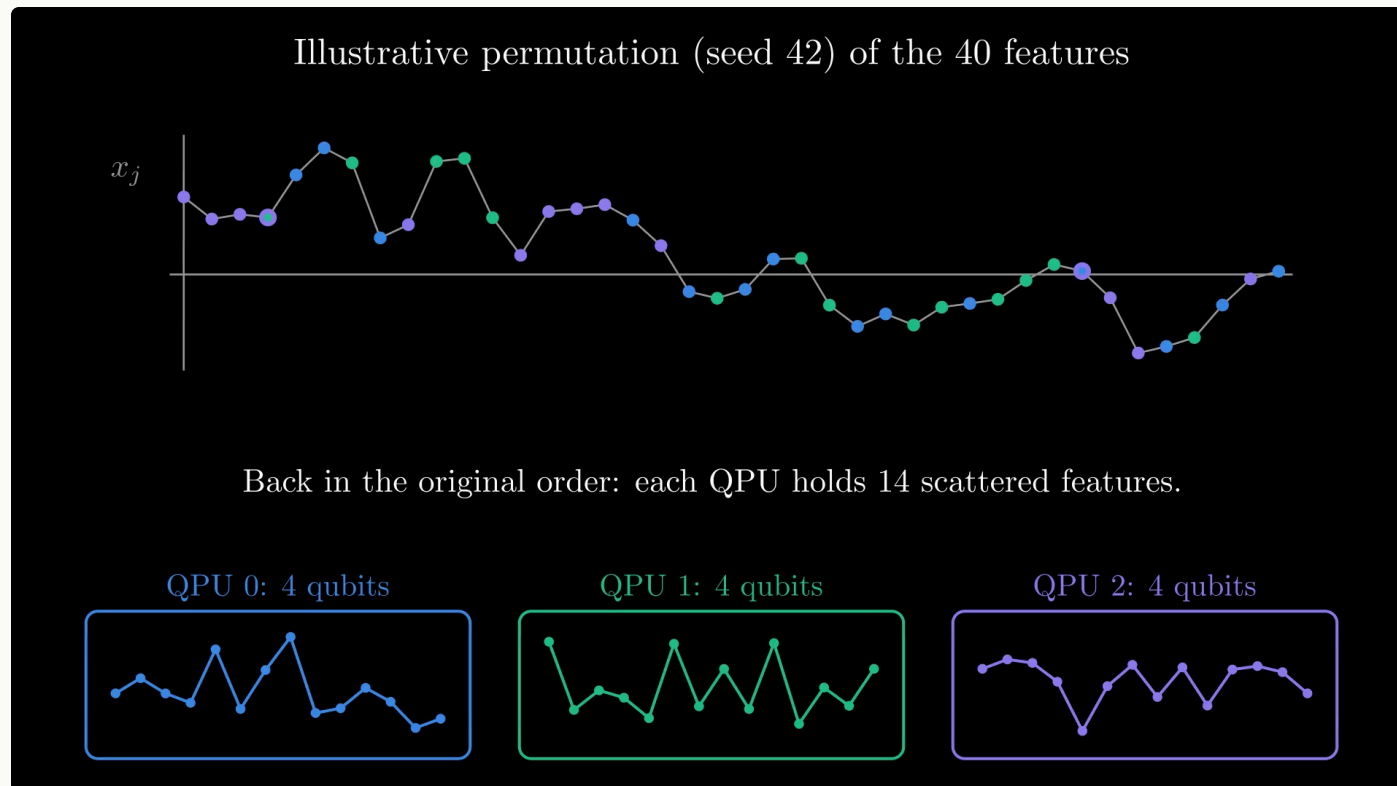


- 3 QPUs $\times n = 4$ qubits, 14 features each, $L = 8$ trainable layers
- Two rounds: each QPU measures one qubit mid-circuit and receives one message bit (dashed: classical only).

METHOD

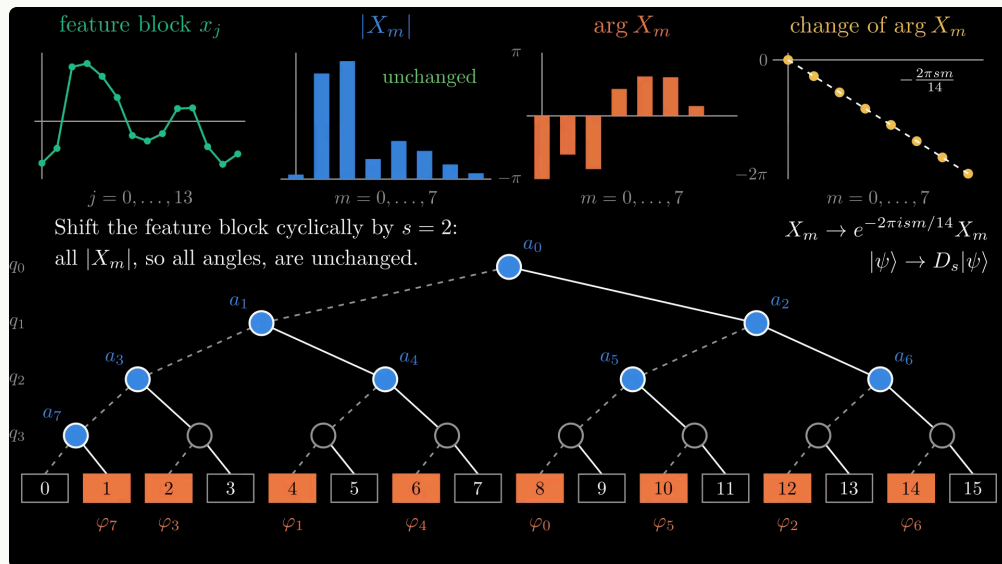
Feature partition: each QPU encodes 14 of the 40 features

- **Contiguous blocks (cyclic):**
0–13, 14–27, 27–39 + 0
- **Permuted features:** same partition after a fixed random permutation
- One block alone: linear classifier on its encoded state ≤ 0.67 (4 digits, chance 0.25)



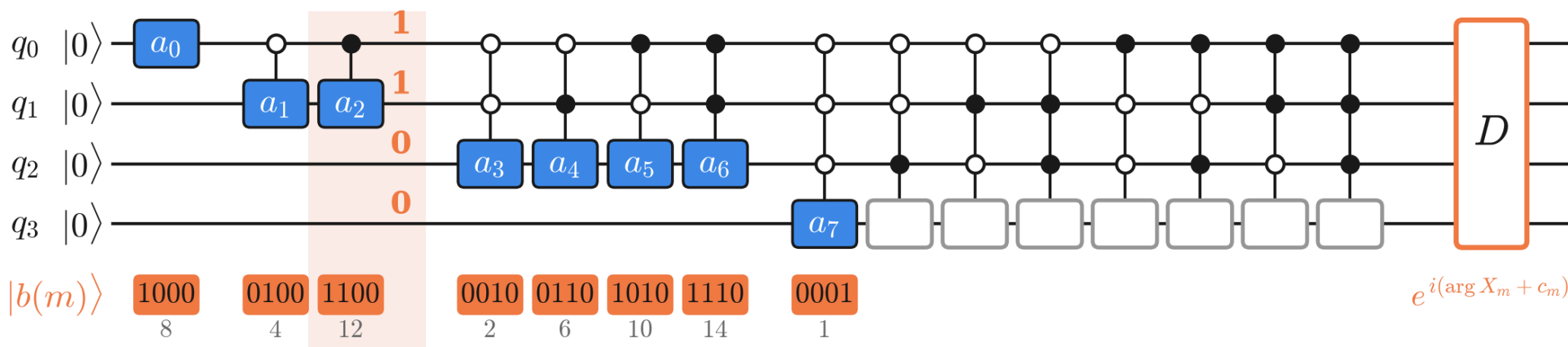
MNIST-1D: Greydanus and Kobak, ICML 2024, arXiv:2011.14439.

The Fourier (DFT) encoding



$$X_m = \sum_{j=0}^{13} x_j e^{-2\pi i j m / 14}, \quad m = 0, \dots, 7$$

- Uniformly controlled R_y : blue
 $a_m = \text{clip}(s_m |X_m| + \beta_m, 0, \pi)$; grey: trainable constants
- Orange $|b(m)\rangle$: a_m 's control values, then 1, then all 0; D adds the phase
- Cyclic shift s : $X_m \rightarrow e^{-2\piism/14} X_m$, one fixed diagonal D_s



Why the choice of phase-carrying basis state matters

- One copy: $P = \text{Tr}[E\rho]$ is linear in $\rho_{xy} = \psi_x \psi_y^*$

$$\rho_{b(m),0} = C_0 A_m \sin(a_m/2) e^{i(\arg X_m + c_m)}$$

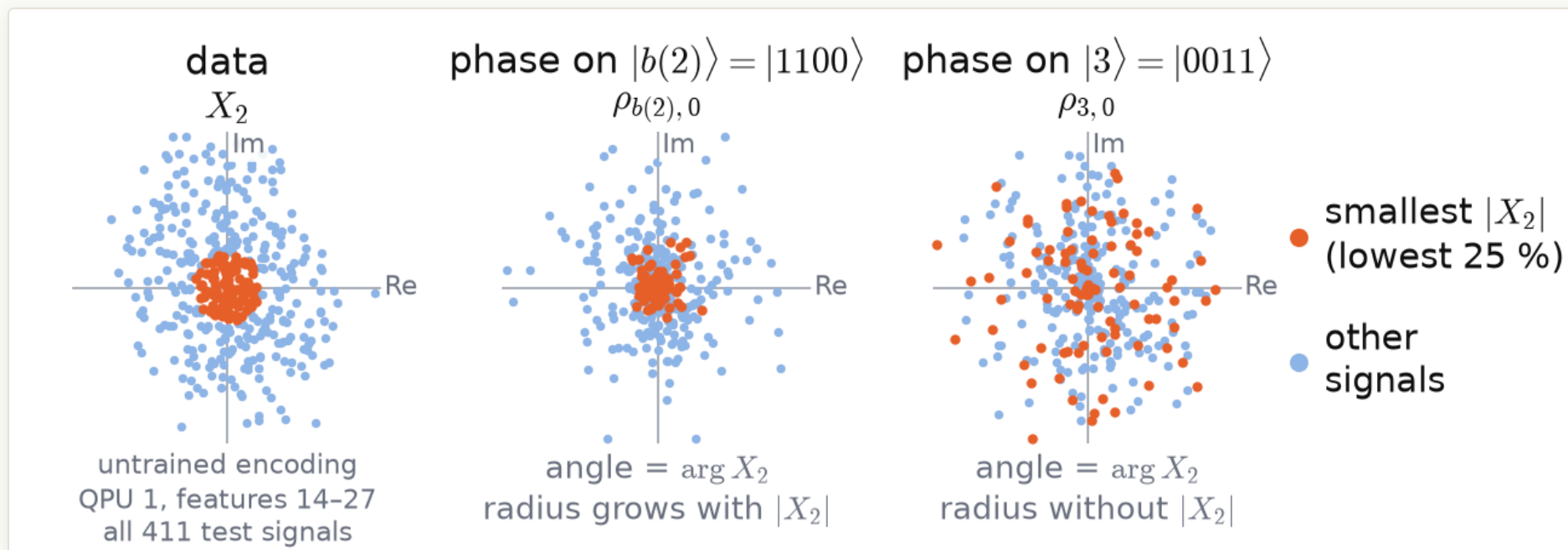
$$C_0 = \psi_{0000} = \prod_{k=0,1,3,7} \cos(a_k/2) \geq 0$$

$$|\psi_{b(m)}| = A_m \sin(a_m/2), A_m \geq 0:$$

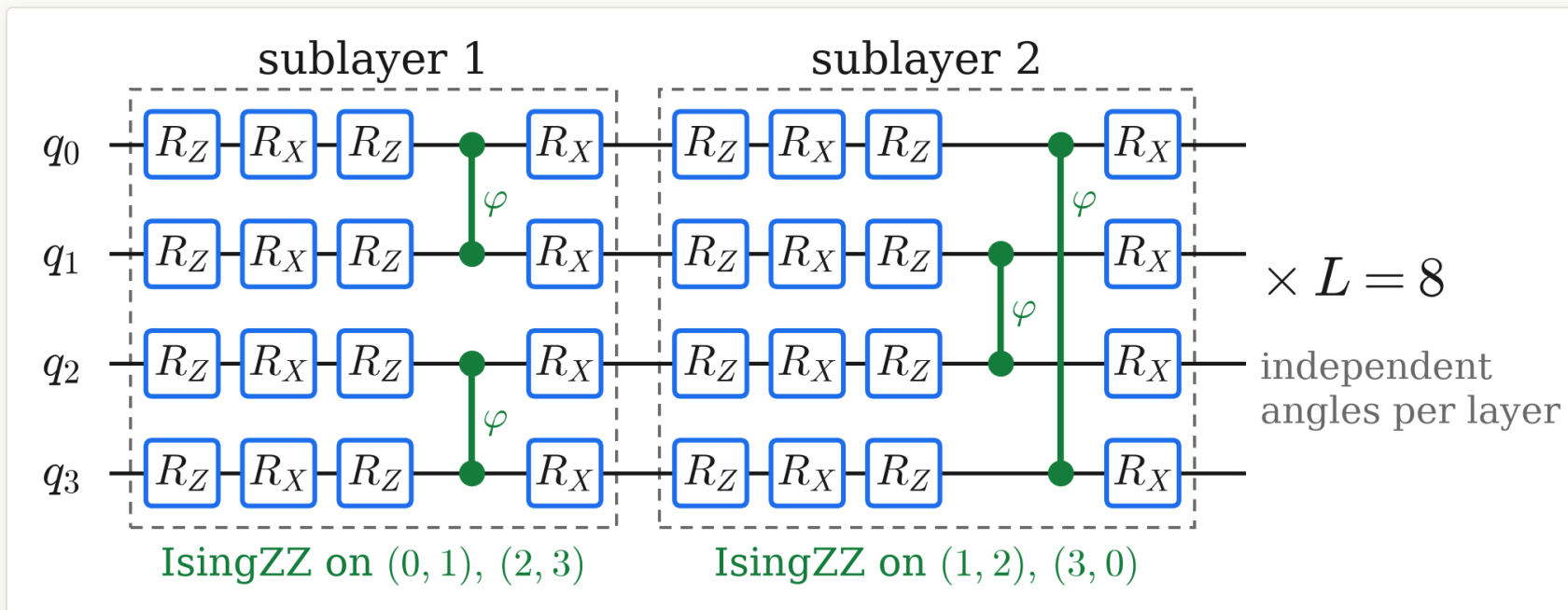
the other cos, sin factors on its path, free of a_m

- Phase on $|b(m)\rangle$:** a rescaled copy of X_m
- Phase on $|m+1\rangle$:** same angle, radius without $|X_m|$

No communication, phase on $|m+1\rangle \rightarrow |b(m)\rangle$:
test accuracy $0.738 \rightarrow 0.865$ (4 digits, seeds 0-2)



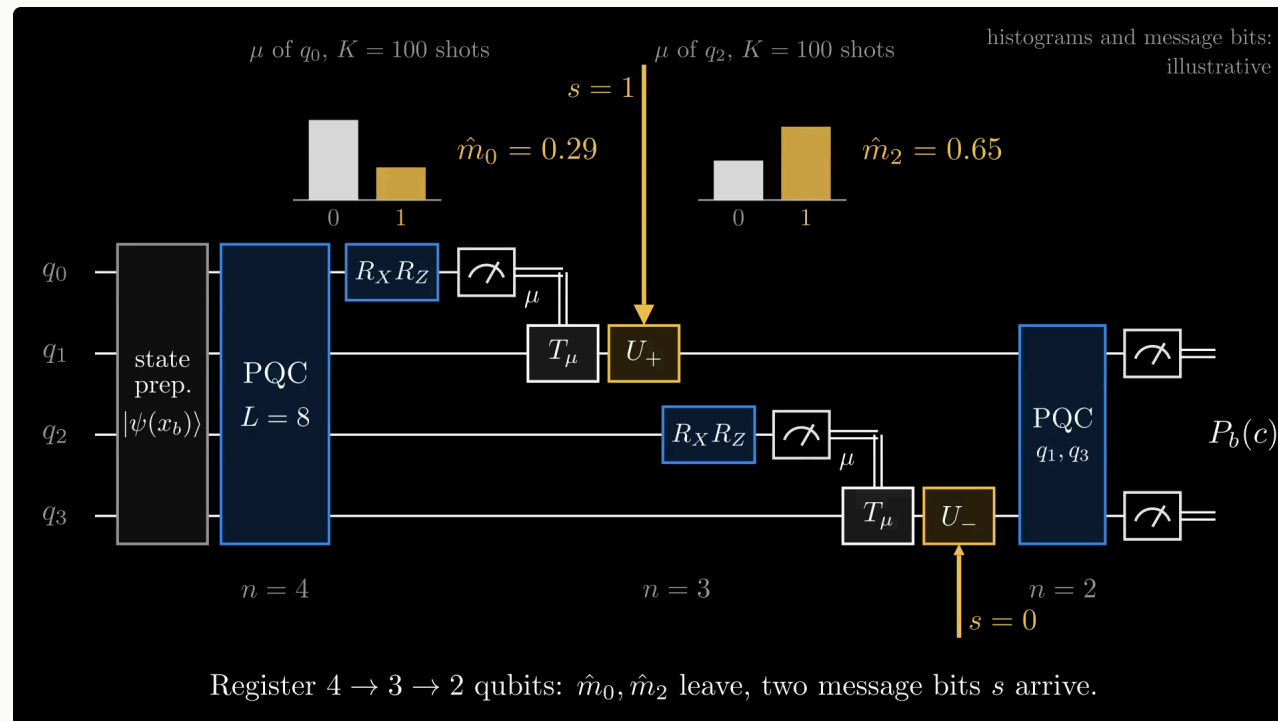
The "convolution" is a parametrised quantum circuit



- $\text{IsingZZ}(\varphi) = e^{-i\varphi Z \otimes Z/2}$: the only entangling gate
- All angles trainable: $8 \times 36 = 288$ of the 434 parameters of a QPU
- Same gate pattern on every neighbouring pair, angles not shared (unlike a CNN kernel or the translation-invariant QCNN layer)

METHOD Pooling is a mid-circuit measurement

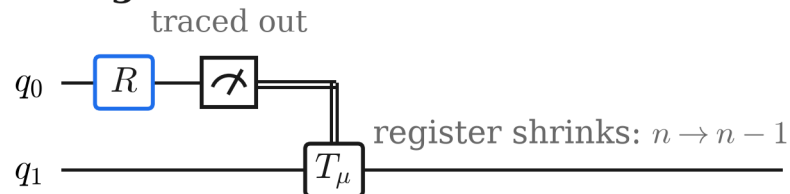
- Each round: measure one qubit mid-circuit, trace it out, apply T_μ to a remaining qubit; $4 \rightarrow 3 \rightarrow 2$ qubits, as in CNN pooling
- K shots give $\hat{m}$; the receiver gets only the message bit s (U_+ or U_-)
- Averaged over μ : still a quantum channel, linear in ρ ; the nonlinearity is the threshold on $\hat{m}$



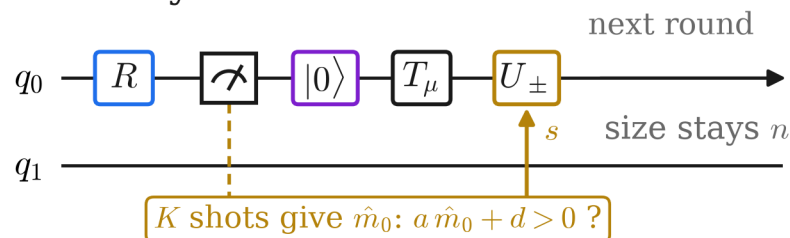
STRATEGY

Qubit reuse: reset the measured qubit instead of tracing it out

Pooling: measure and trace out



Reuse: measure, reset, then a classically controlled rotation



- Size stays n , so the number of rounds R is free
- A threshold on the QPU's own $\hat{m}_0$ needs many copies: a nonlinearity no single-copy circuit has

phase on, R	local – no control	two-QPU (CC) – local
$ m+1\rangle, 2$	$+2.6 \pm 2.7$	$+9.7 \pm 3.0$
$ m+1\rangle, 4$	$+9.7 \pm 4.7$	$+6.5 \pm 4.8$
$ b(m)\rangle, 2$	$+0.1 \pm 1.0$	$+1.7 \pm 1.0$

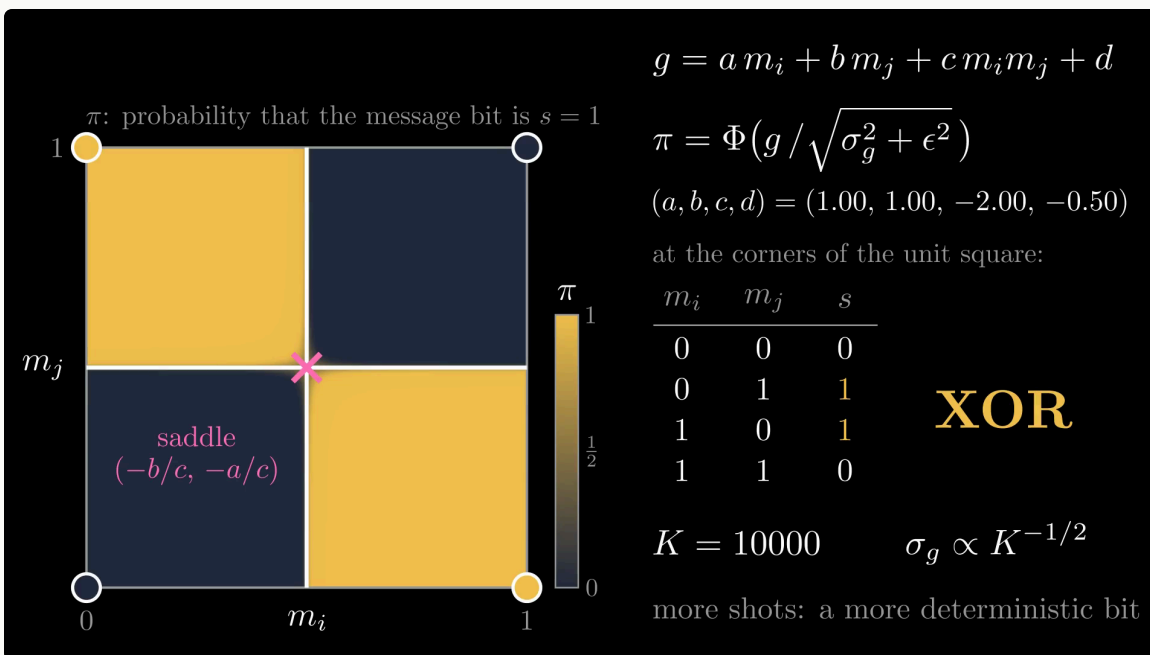
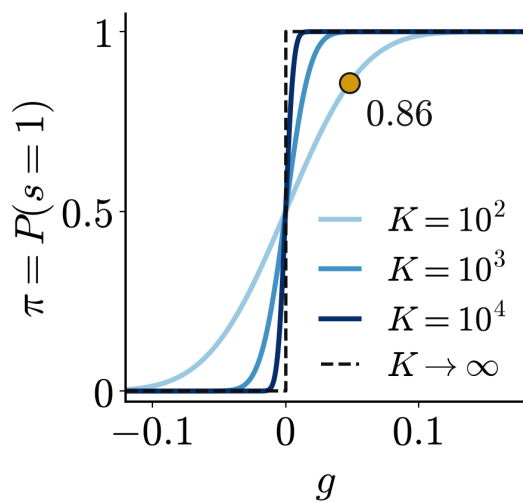
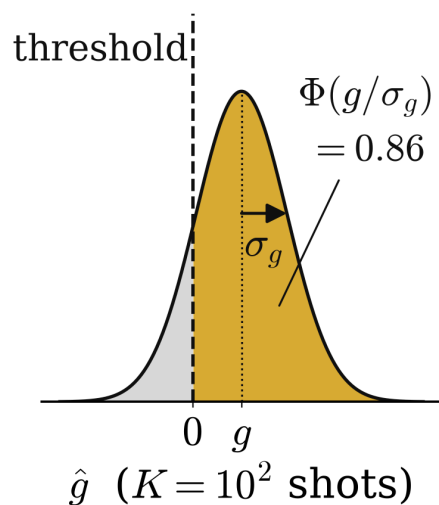
- Phase on $|b(m)\rangle$: gain within noise

Test-accuracy differences (percentage points), 4 digits, $n = 4$, contiguous blocks, 5-layer circuit per round; $\pm$ Welch standard error, 3 or 5 seeds.

The decision function: a stochastic threshold (probit)

$$g = a m_i + b m_j + c m_i m_j + d$$

$$P(s=1) = \pi = \Phi\left(g / \sqrt{\sigma_g^2 + \epsilon^2}\right)$$

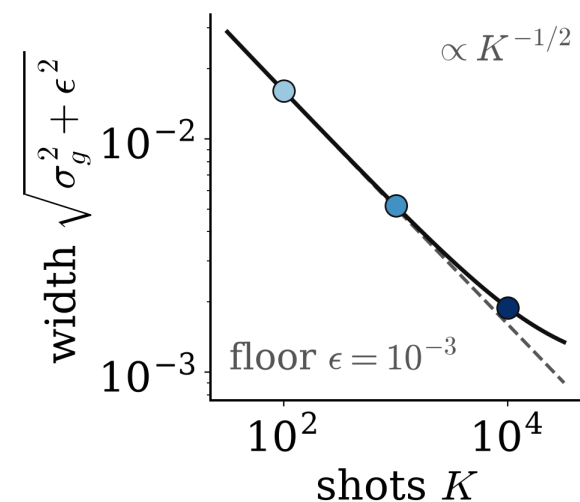
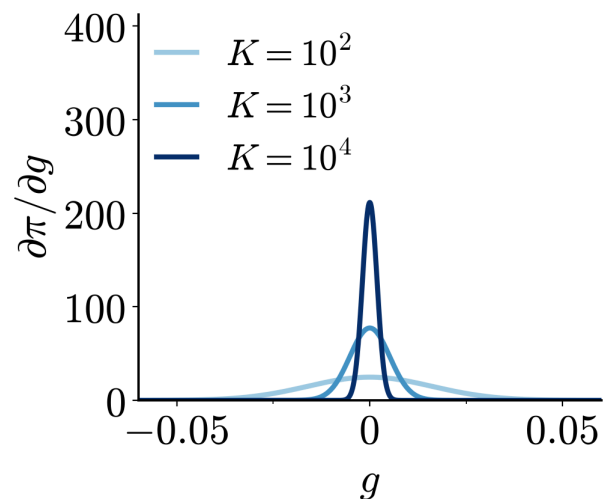


- m_i, m_j : outcome-1 probabilities from K shots; shot noise $\sigma_g \propto K^{-1/2}$
- Φ : standard normal CDF; $s = 1$ applies U_+ , else U_-
- Corners $\{0, 1\}^2$: the sign of g is a Boolean function (AND, OR, XOR)

STRATEGY (a, b, c, d) are now trainable

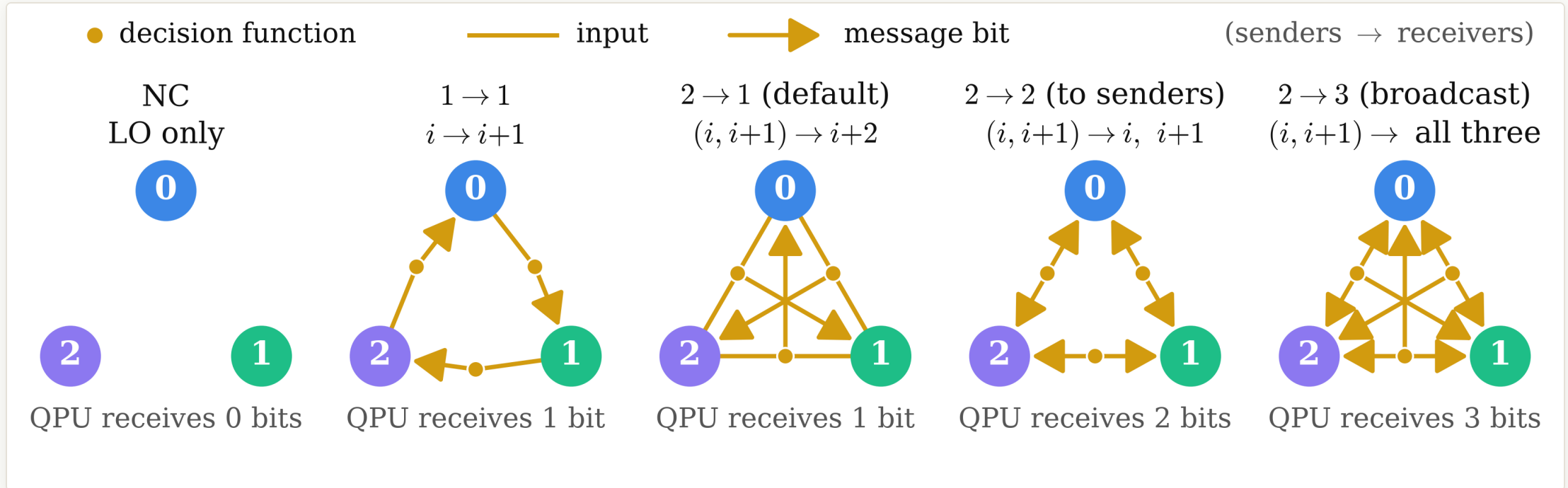
	CY's code (Feb 2026)	now
input m	single-shot bits $\{0, 1\}$	K -shot estimates $[0, 1]$
message bit s	deterministic threshold	stochastic: $P(s=1) = \pi(g)$
(a, b, c, d)	fixed, or SPSA	backpropagation (all six g)

- Exact gradient $\partial\pi/\partial g = \varphi(g/\sigma)/\sigma$: appreciable only for $|g| \lesssim \sigma \propto K^{-1/2}$
- Shot schedule $K = 10^2 \rightarrow 10^4$: +4.5 percentage points over fixed $K = 10^4$



SPSA (simultaneous perturbation stochastic approximation): J. C. Spall, IEEE Trans. Autom. Control 37, 332 (1992).

Communication topologies compared: three messages per round

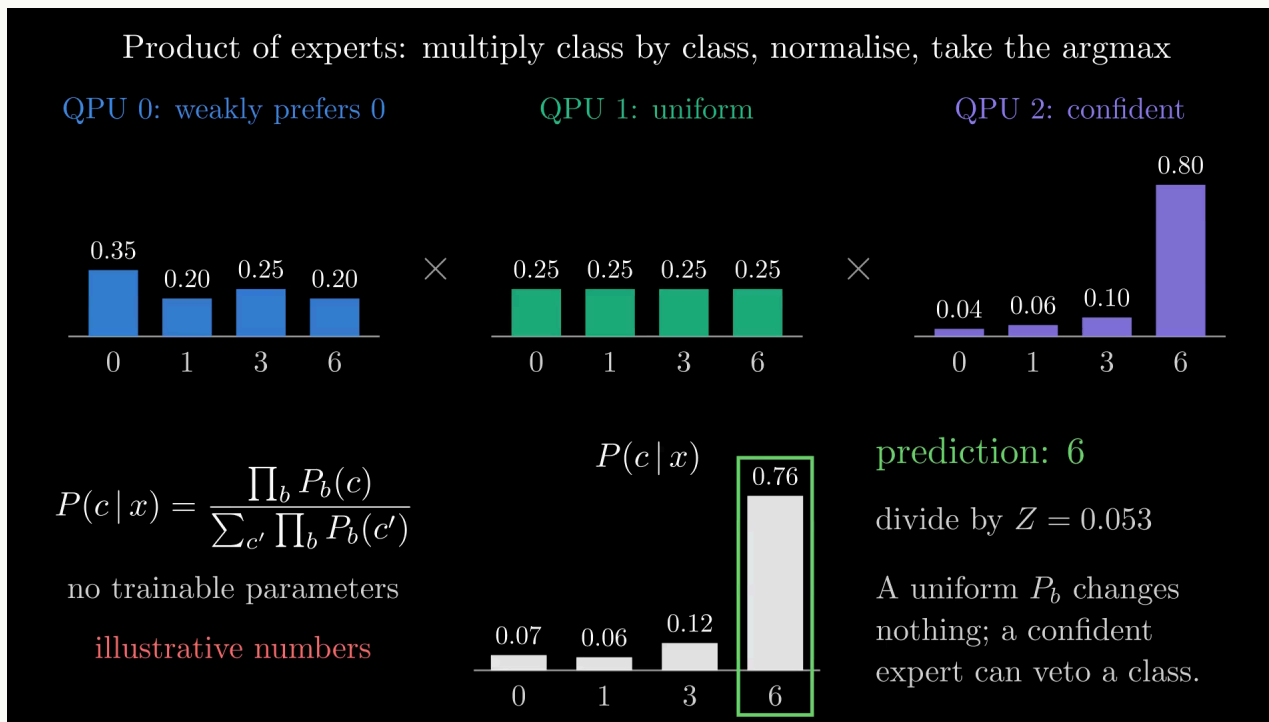


- NC (no communication): local operations only (LO); the other four are LOCC, two rounds
- $1 \rightarrow 1$ reads one QPU: $g = a m_i + d$
- $2 \rightarrow 1, 2 \rightarrow 2, 2 \rightarrow 3$ compute the same three $g(m_i, m_{i+1})$; only the receivers differ

How the prediction is made: a product of experts

$$P(c|x) = \frac{\prod_b P_b(c)}{\sum_{c'} \prod_b P_b(c')}$$

- $P_b(c)$: two output qubits of QPU b , one outcome per digit
- Uniform P_b : no effect; confident P_b : veto
- No parameters; without communication: naive Bayes

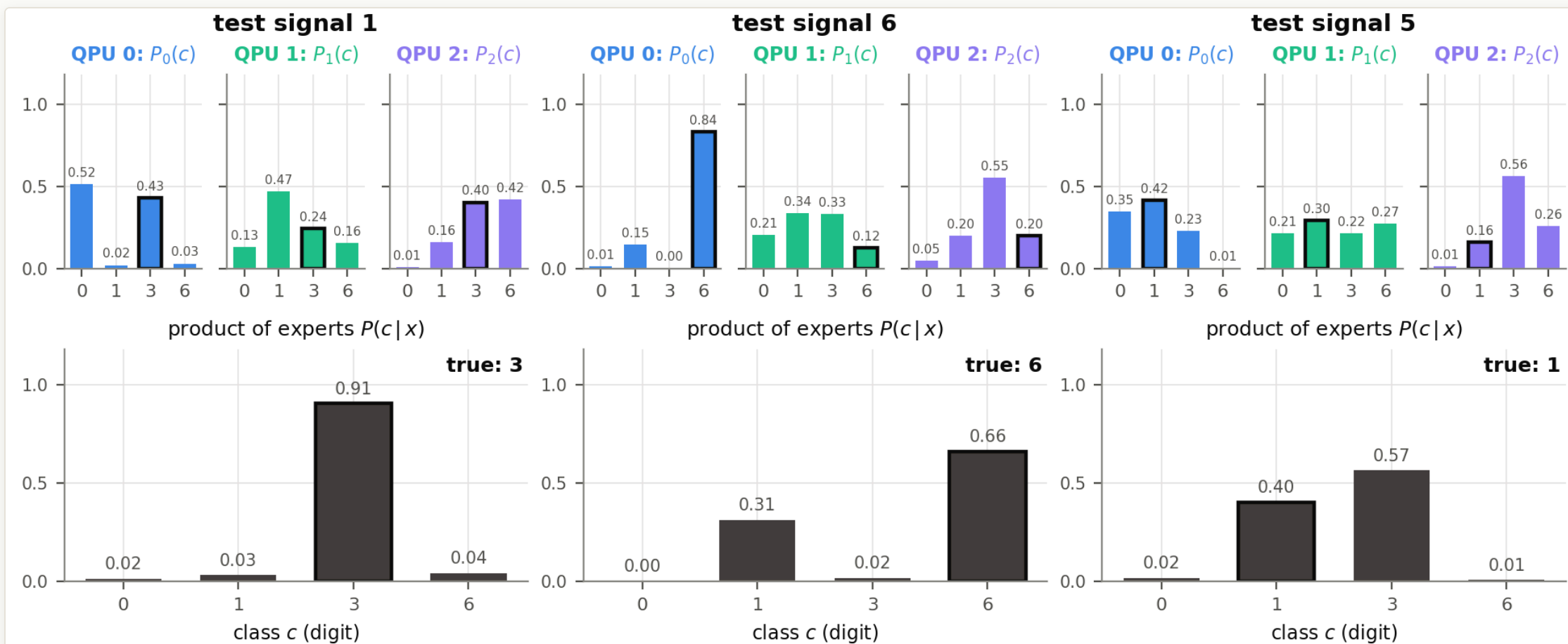


Product of experts: G. E. Hinton, Neural Comput. 14, 1771 (2002).

RESULTS

The trained model on three test signals

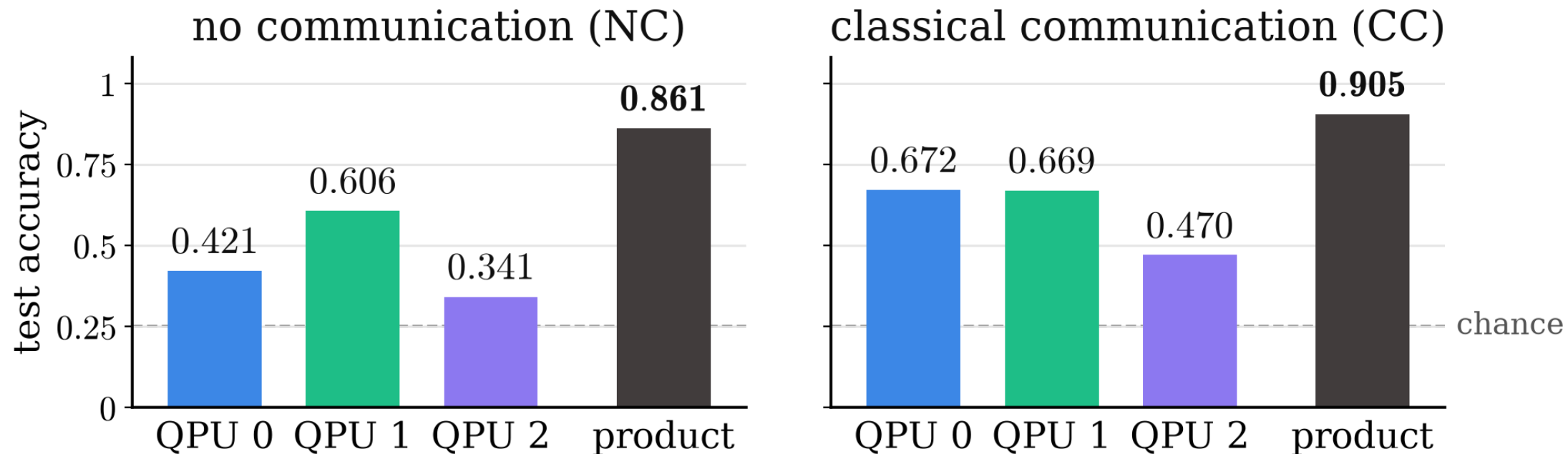
- Trained model with classical communication (CC), seed 0; top: each QPU's $P_b(c)$, bottom: the product
- Signal 1: every QPU alone is wrong; the product gives digit 3 probability 0.91
- Signal 6: QPU 0 vetoes 1 and 3; signal 5: one confident wrong QPU outvotes two



RESULTS

The product is right more often than any QPU alone

4 digits, $n = 4$ qubits per QPU, seed 0 (one run)



- **Rescues:** correct signals with at least two QPUs wrong alone: 196 of 354 (NC), 82 of 372 (CC)
- **With communication,** QPUs 0 and 2 improve most: the message bits bring them a summary of feature block 1

Training: cross-entropy loss and the training setup

$$L = -\frac{1}{B} \sum_{i=1}^B \log P(y_i | x_i)$$

- Uniform guess (four digits):

$$\ln 4 \approx 1.386 \text{ nats}$$

- Product-of-experts readout:

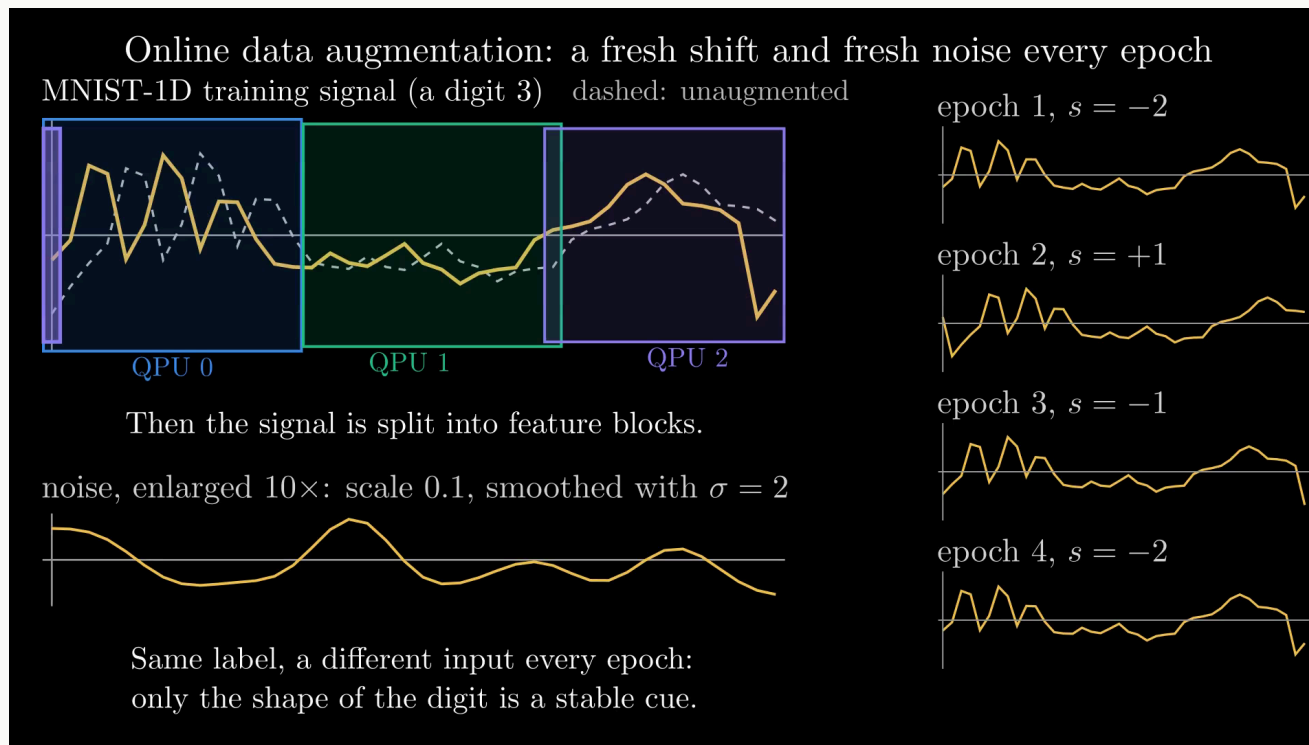
$$\log P(y | x) = \sum_b \log P_b(y) - \log Z(x)$$

step	choice
optimizer	Adam, learning rate 0.03, cosine decay
length	200 epochs, batch $B = 256$
shots	$K: 10^2 \rightarrow 10^4$, geometric
model selection	best epoch on 264 validation signals; 411 test signals
simulation	exact density matrices
compute	about 900 runs, Colab (A100 GPU, CPU)

Adam: D. P. Kingma and J. Ba, ICLR 2015, arXiv:1412.6980.

Online data augmentation: a freshly augmented sample every epoch

- Each epoch: circular shift $|s| \leq 2$ + correlated noise 0.1
- Generalization gap ($n = 4$): 0.008 without communication, 0.035 with
- Random message bits (same mean) remove about half of the 0.035



Gaps: seeds 3–7, not used for model selection. Augmentation as in the MNIST-1D generator: Greydanus and Kobak, ICML 2024, arXiv:2011.14439.

05

Results

Accuracy, and what communication adds

RESULTS

Accuracy: just below the 0.9 target

Four digits (0, 1, 3, 6), 411 test signals, three QPUs of $n = 4$ qubits; mean $\pm$ s.d. over seeds 3–7 (not used for model selection).

model	test accuracy	generalization gap
no communication (NC)	0.878 ± 0.007	0.008
classical communication (CC), 2 senders $\rightarrow$ 1 receiver	0.890 ± 0.008	0.035
pre-registered target	> 0.9	≤ 0.03
classical, all 40 features: tuned RBF SVM / kernel logistic regression (shift-aware)	0.964 / 0.978	

- **Target not met:** 1828 of 2055 test predictions correct, 1850 needed.
- **The gap comes with communication:** random message bits remove about half of it.
- Seeds 0–2 were used to choose the encoding (CC there: 0.908).

RESULTS

Most of the gain came from the Fourier (DFT) encoding

Block DFT $X_m = |X_m|e^{i\varphi_m}$; a cyclic shift only moves phases: $X_m \rightarrow e^{-2\piism/14}X_m$.

NC / CC: no / classical communication ($2 \rightarrow 1$); 4 digits, seeds 0–2, mean $\pm$ s.d.

encoding	state (a_m : tree angle set by $ X_m $)	linear classifier (untrained)	trained, NC	trained, CC
(1) tree state, phase on $ m+1\rangle$	$\psi_{m+1} \propto \sin \frac{\theta_t}{2} e^{i\varphi_m},$ θ_t : another node's angle	0.773	0.738 ± 0.014	0.848 ± 0.027
(2) tree state, phase on $ b(m)\rangle$	$\psi_{b(m)} \propto \sin \frac{a_m}{2} e^{i\varphi_m}$	0.943	0.865 ± 0.014	0.908 ± 0.002

- **(2) vs (1):** $+12.7 \pm 1.1$ points without, $+5.9 \pm 1.6$ with CC (three of three seeds).
- **Less underfitting:** training accuracy rises as much; NC gap $4.1 \rightarrow 1.6$ points.

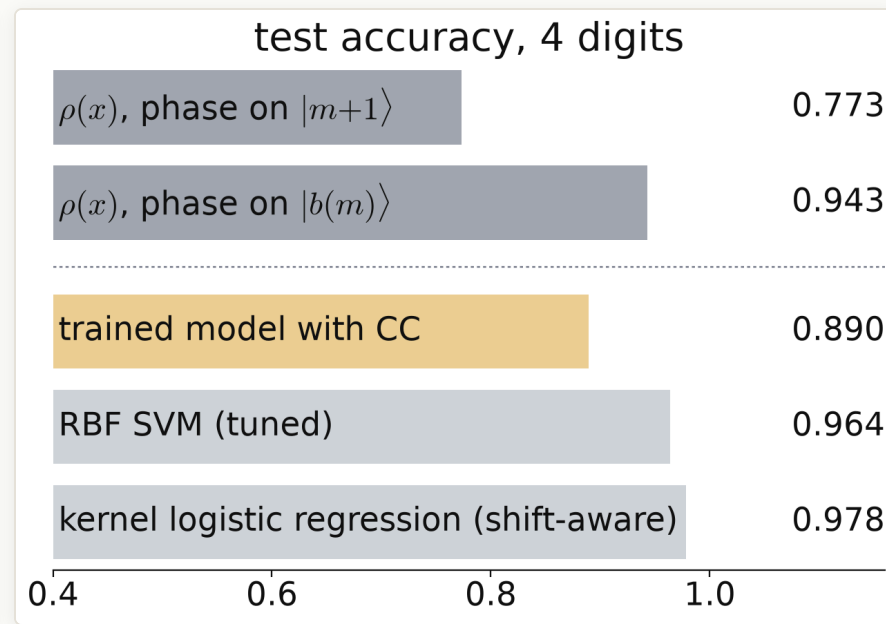
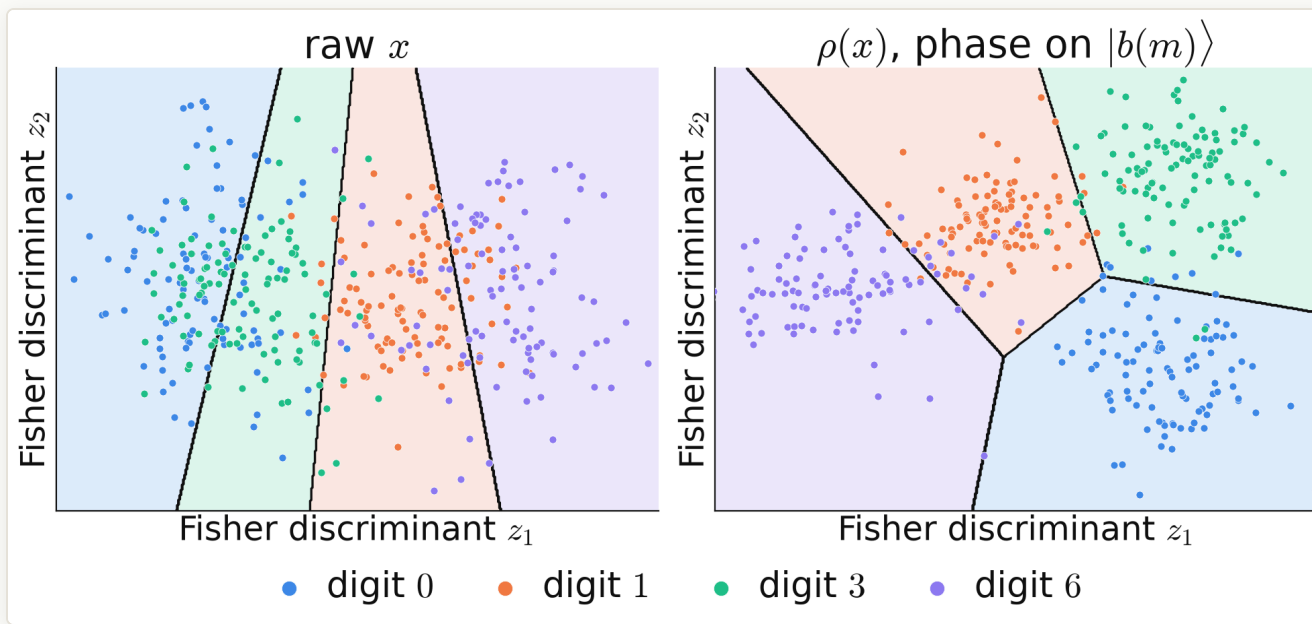
RESULTS

A linear classifier on the encoded states already reaches 0.943

Logistic regression on the entries of the three encoded states $\rho_b(x)$:

$$\text{score}_c(x) = \sum_{b=0}^2 \text{Tr}[W_c^{(b)} \rho_b(x)] + \beta_c, \quad \hat{c} = \arg \max_c \text{score}_c.$$

A QPU without incoming messages, $P_b(c) = \text{Tr}[E_c \rho_b]$ (POVM elements E_c), is such a classifier: this accuracy bounds it.



RESULTS

A monolithic QPU does worse (still trying)

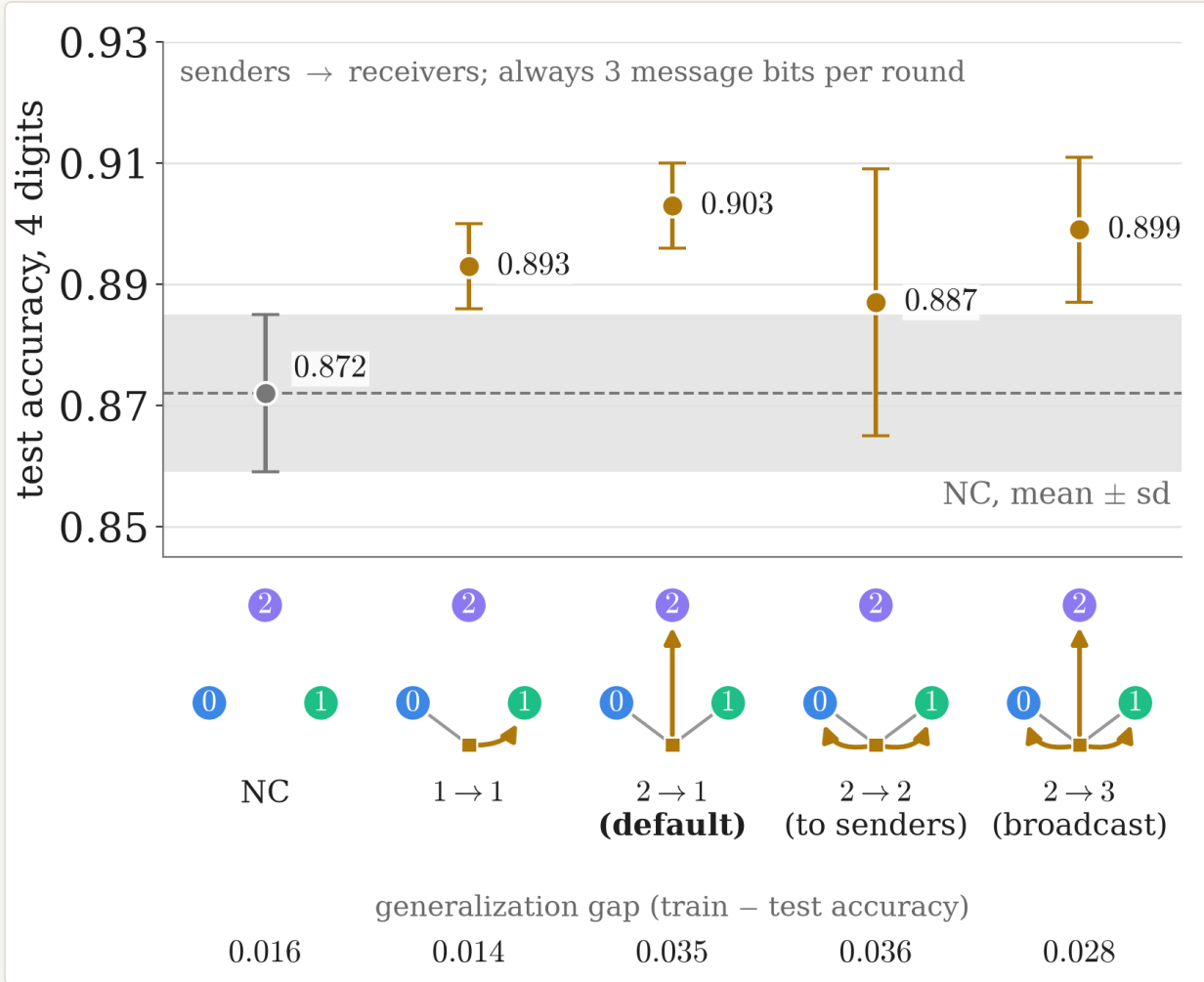
model (phase on $ b(m)\rangle$, 4 digits, three seeds)	circuit parameters	test accuracy
monolithic 12-qubit QCNN	1656	0.805 ± 0.076
three 4-qubit QPUs, no communication (seeds 0–2)	1128	0.865
three 4-qubit QPUs, classical communication (seeds 0–2)	1188	0.908

- **Loses on all three seeds**, despite more parameters and entanglement across the feature blocks.
- **Its trained encoded states are worse**: linear classifier 0.864 vs 0.946.
- **Not a matched baseline**: circuit, pooling and readout all differ.

QCNN (quantum convolutional neural network): I. Cong, S. Choi and M. D. Lukin, Nature Physics 15, 1273 (2019).

RESULTS

Communication topologies: every one is a good model



- Seeds 0–4: every topology beats no communication (NC) by 1.5 to 3.1 points; they differ by at most 1.6, within the seed spread.
- **Two senders vs one:** 77 % vs 23 % non-constant decision functions; $+0.9 \pm 0.5$ points ($\pm$ standard error), a trend.
- **More receivers do not help:** $2 \rightarrow 2$ and $2 \rightarrow 3$ are no better.

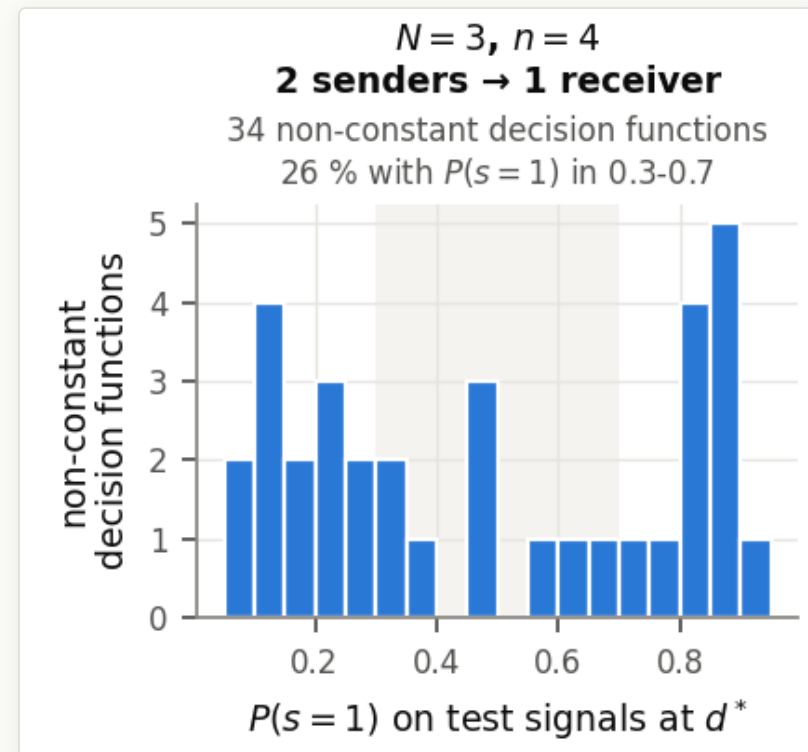
RESULTS

What the trained decision functions compute

6 decision functions per model $\times$ seeds 0–7 = 48, read at the corners $\{0, 1\}^2$:

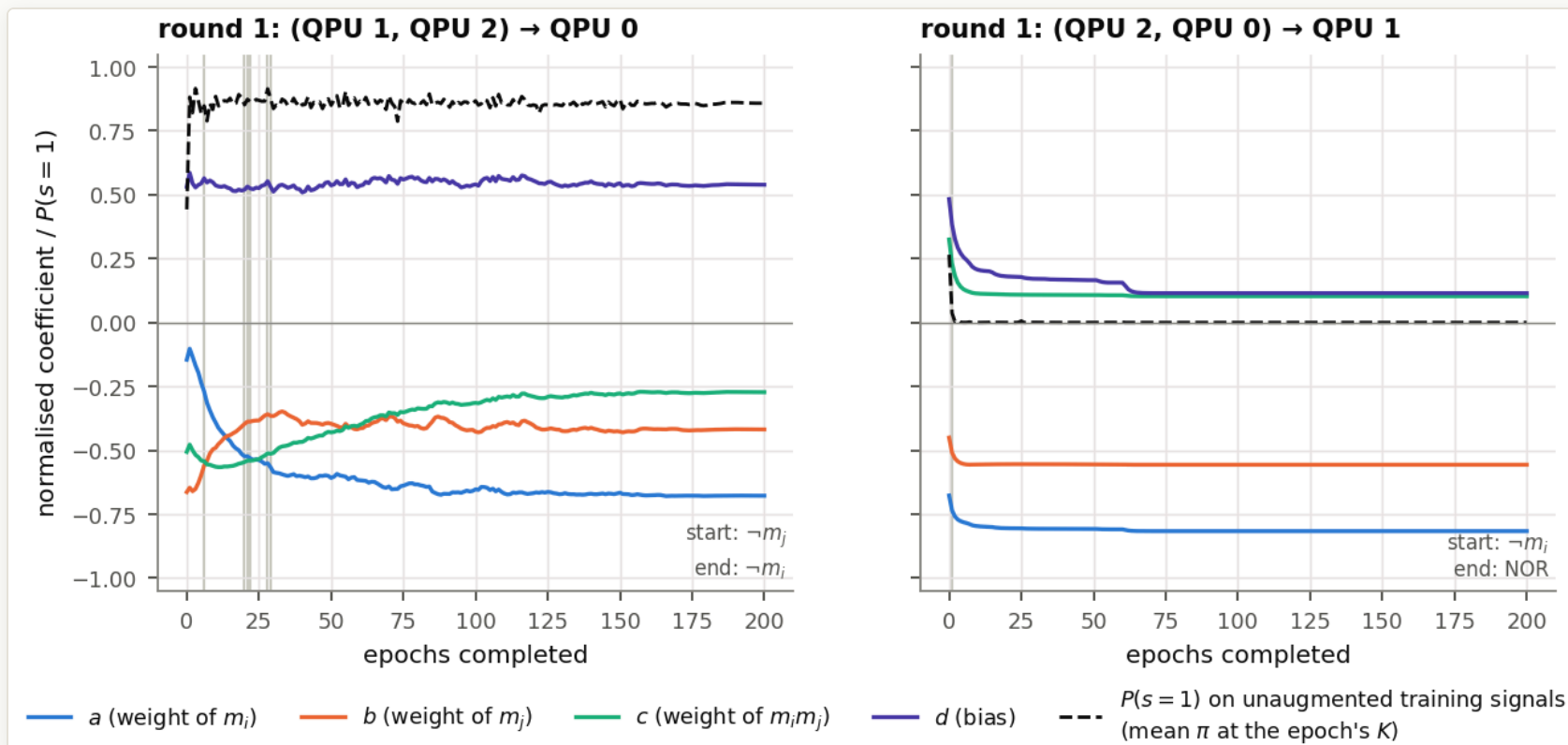
Boolean function	count
constant	14
dictator or its negation ($m_i, \neg m_j, \dots$)	20
implication and its variants ($\neg m_i \vee m_j, \dots$)	9
AND, NAND, NOR	5
XOR	0

- Many ignore an input: constant, or one QPU's estimate only. Task or trainability? Open.
- The 34 non-constant ones are mostly unbalanced: only 26 % have $P(s = 1)$ in 0.3–0.7; the bit flags a subgroup of signals.



RESULTS

The coefficients move early, then settle

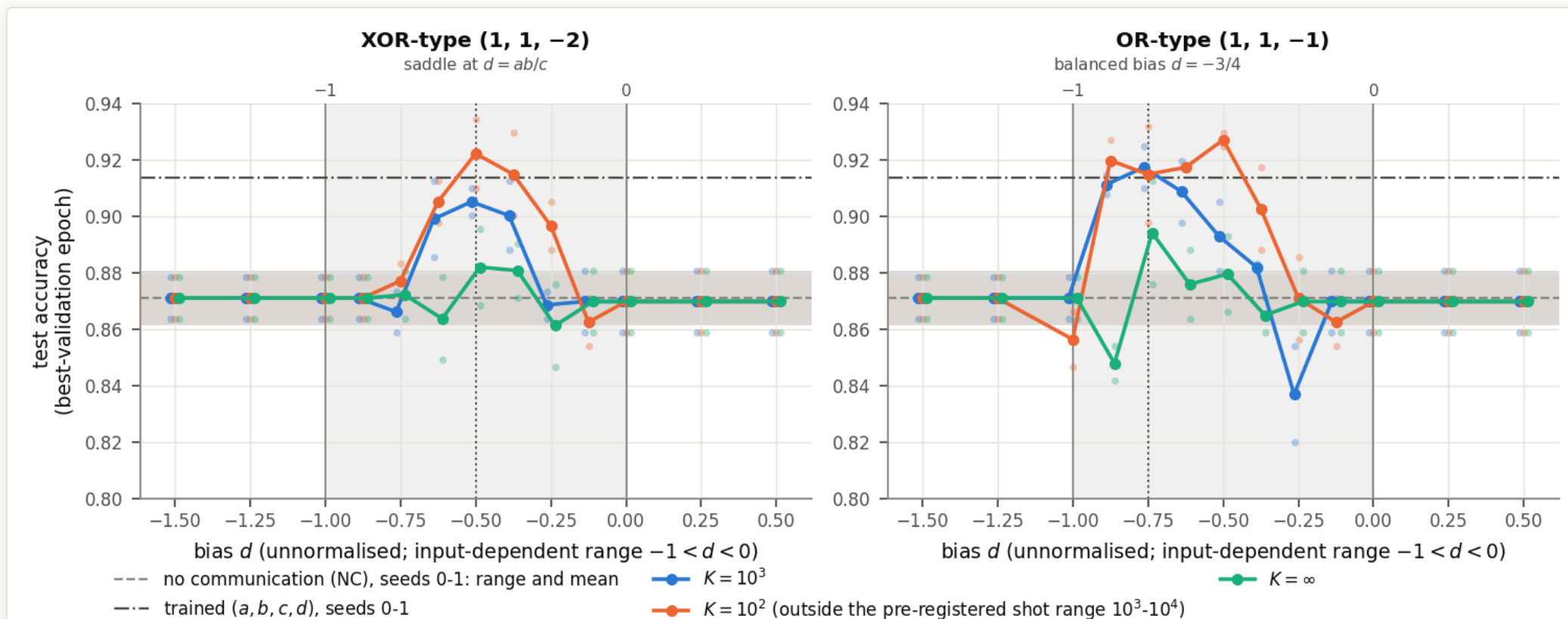


Seed 0 (test accuracy 0.905, 4 digits):

- Four of the six decision functions keep their initial Boolean function; shown are the two that change.
- Left: switches (grey lines), ends as $\neg m_i$. Right: becomes NOR, then constant.
- Trained, but not dramatically (one seed).

RESULTS

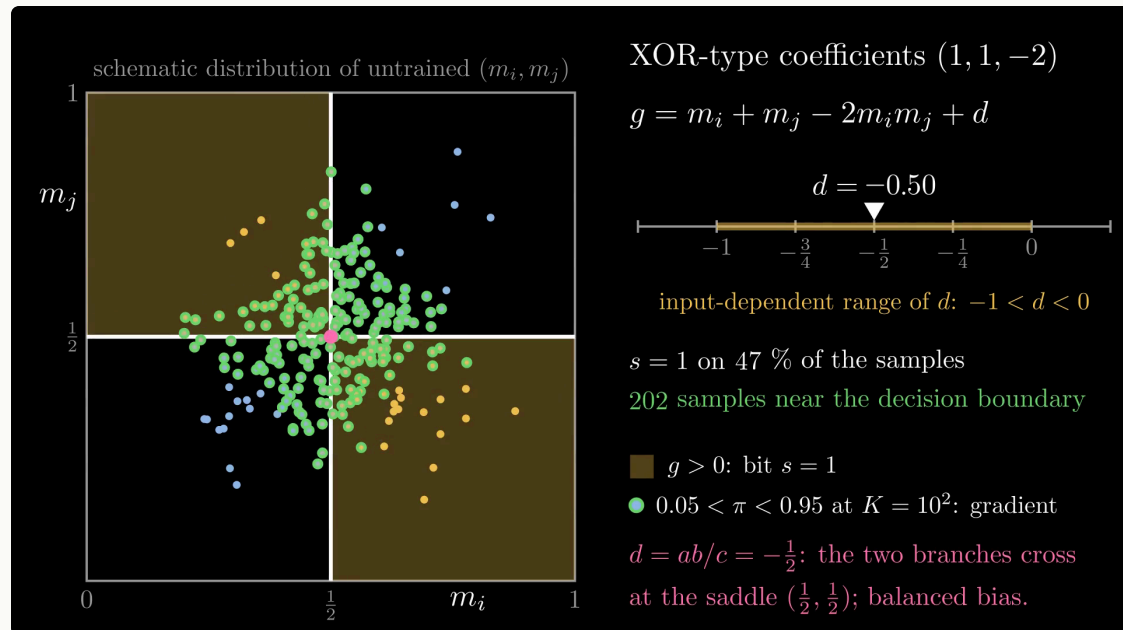
Fixing (a, b, c) and sweeping the bias d



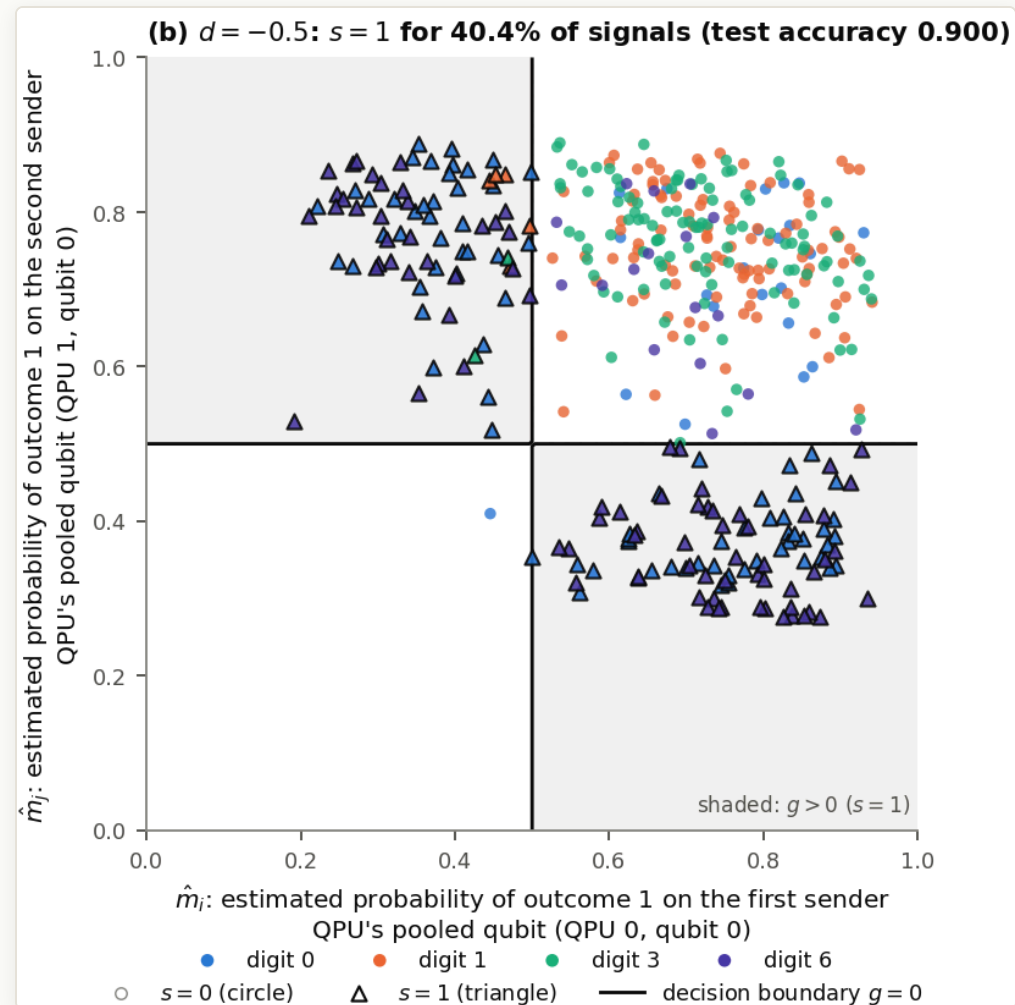
- Outside the input-dependent range $-1 < d < 0$ the bit is constant: accuracy equals no communication (NC).
- Training succeeds only near the balanced bias: $d = -\frac{1}{2}$ (XOR-type), $d = -\frac{3}{4}$ (OR-type).
- An optimal d exists: OR-type at $d = -0.75$ gives 0.917 (4 digits), equal to trained coefficients (0.914).

RESULTS

Only a decision boundary that intersects the data can learn



- XOR-type, $d = -0.875$: boundary outside the data, $s = 0$ always; no gain (accuracy 0.878, 4 digits).
- XOR-type, $d = -0.5$ (right): training forms clusters; the bit asks "0 or 6, not 1 or 3?" (accuracy 0.900).
- Gradient only where $|g| \lesssim \sigma_g$ (shot-noise width): a boundary outside the data never learns.



06

Skepticism and a suggestion

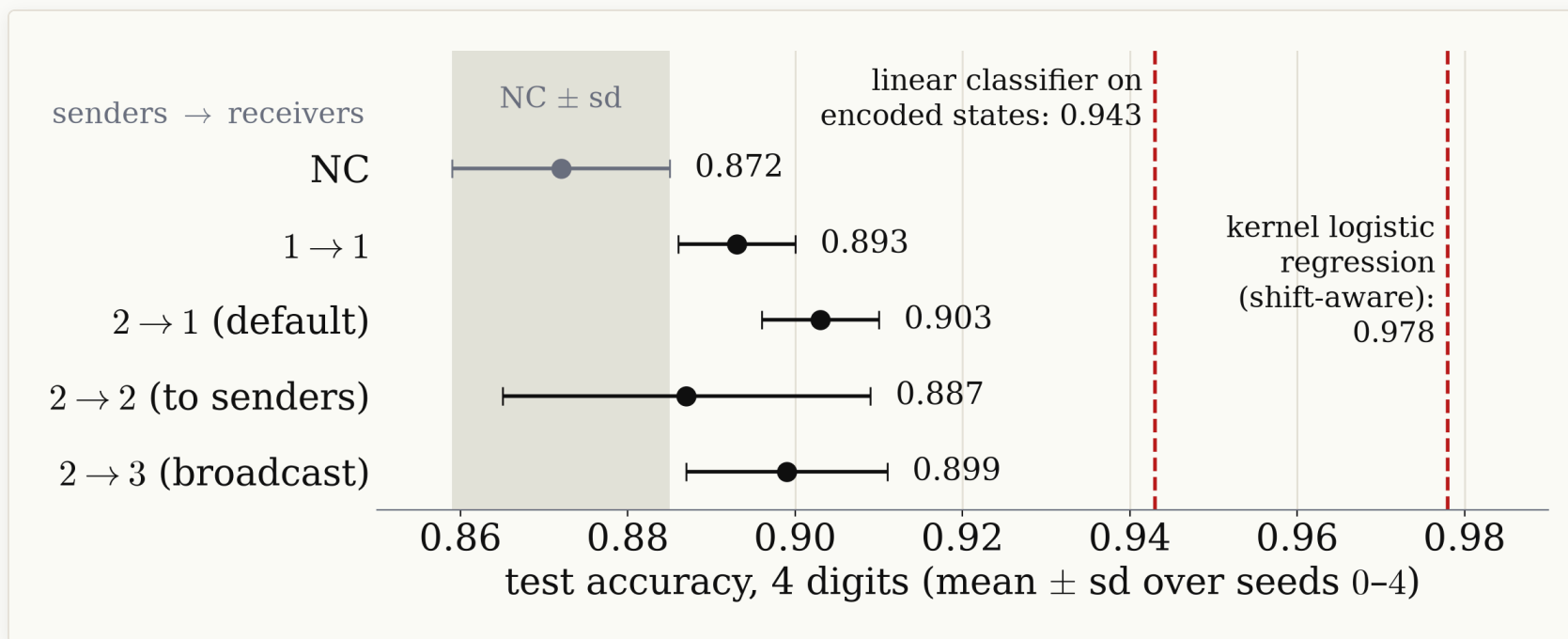
A task where communication must matter

Doubt 1: a feature partition is not an inherently distributed task

classical model, test accuracy, 4 digits	contiguous blocks	permuted features
best additive (per-block) model	0.971	0.891
below the best joint model, kernel logistic regression (shift-aware): 0.978 on all 40 features	0.7 points	8.7 points

- **Contiguous:** class information lies almost entirely within blocks; still, communication adds 1.1 to 3.1 points to our model (by seed set).
- **Permuted:** 8.7 points need combinations across blocks, but the Fourier (DFT) encoding loses its motivation.

Doubt 2: the Fourier (DFT) encoding is too good to reveal the topology



- Topologies differ by at most 1.6 points, about the seed spread.
- Every model stays below the linear classifier on the untrained encoded states (0.943): the encoding did most of the work.

Doubt 3: do decision functions keep their initial Boolean function?

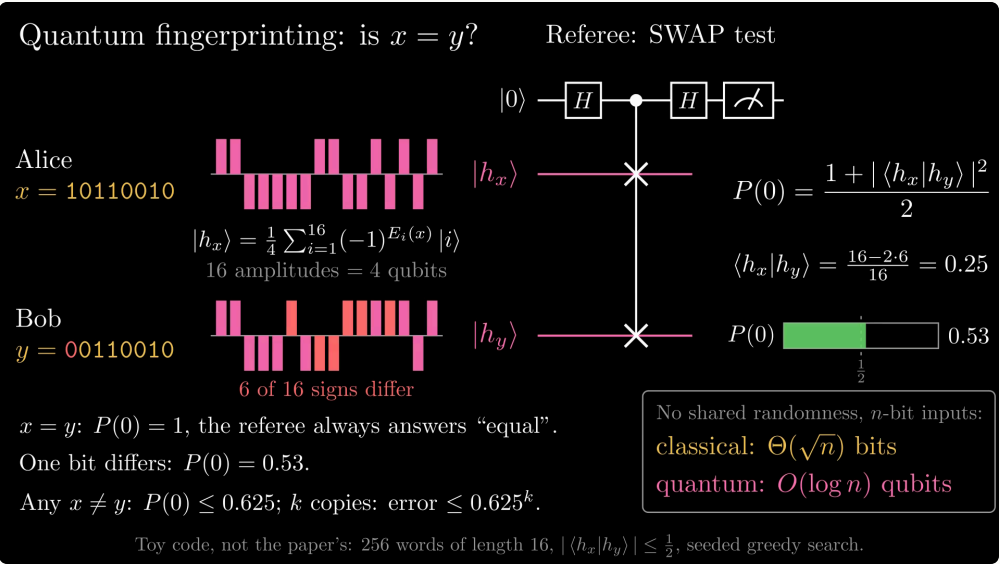
- **Observed.** Seed 0: 4 of 6 keep their initial function. Seeds 0–7: 14 of 48 constant, 20 depend on one input.
- **Possible cause.** Only a boundary that intersects the data gets a gradient: training succeeds only for d from -0.625 to -0.25 (XOR-type) or -0.875 to -0.375 (OR-type).
- **Tried.** Re-initializing d at the data median (phase on $|m+1\rangle$): 94–100 % become input-dependent; accuracy within noise.

So trainability, not usefulness, may choose the Boolean function.

STRATEGY

Suggestion: a distributed task, not a classification benchmark

	now: MNIST- 1D (4 digits)	proposal: inherently distributed task
each QPU holds	a contiguous block of one signal	senders: x or y ; receiver: none
label	mostly local (additive model: 0.971)	a joint function $f(x, y)$: equality, Hamming distance, disjointness, ...
communication	adds 1.1 to 3.1 points	necessary
figure of merit	test accuracy	message size at a given worst-case error
quantum advantage	none observed: ours ≈ 0.90 , kernel logistic regression (shift-aware) 0.978	proven for some f , not for all



Goal: an answer that depends jointly on inputs no single QPU holds.

Example, equality: each sender sends a quantum fingerprint, the referee runs a SWAP test.

Buhrman, Cleve, Watrous, de Wolf, Phys. Rev. Lett. 87, 167902 (2001).

STRATEGY

Joint functions $f(x, y)$: quantum vs classical communication

$f(x, y)$	model (bounded error)	qubits	bits	reference
equality $[x = y]$	simultaneous (SMP), no shared randomness	$O(\log n)$	$\Theta(\sqrt{n})$	Buhrman et al. 2001
Hamming distance $\leq d$ (d fixed)	SMP, no shared randomness	$O(\log n)$	$\Theta(\sqrt{n})$	Yao 2003
disjointness $[x \cap y = \emptyset]$	two-way	$\Theta(\sqrt{n})$	$\Theta(n)$	Aaronson, Ambainis 2003
Boolean hidden matching, variant (promise)	one-way	$O(\log n)$	$\Theta(\sqrt{n})$	Gavinsky et al. 2007
vector in subspace (promise)	quantum one-way, classical two-way	$O(\log n)$	$\Omega(n^{1/3})$	Klartag, Regev 2011
inner product mod 2: no advantage	two-way, even with entanglement	$\Theta(n)$	$\Theta(n)$	Cleve et al. 1998

- **Our architecture:** the senders hold x and y ; the receiving QPU, given no input, is the SMP referee.
- **First target, equality:** SMP matches our 2 senders $\rightarrow$ 1 receiver, and the referee needs only a SWAP test.
- **Measure** worst-case error vs message size, CC vs QC (trained parameters act as a shared key).

Buhrman, Cleve, Watrous, de Wolf, PRL 87, 167902 (2001); Yao, STOC 2003; Buhrman, Cleve, Wigderson, STOC 1998; Aaronson, Ambainis, Theory Comput. 1, 47 (2005); Razborov, Izv. Math. 67, 145 (2003); Bar-Yossef, Jayram, Kerenidis, STOC 2004; Gavinsky, Kempe, Kerenidis, Raz, de Wolf, STOC 2007; Klartag, Regev, STOC 2011; Cleve, van Dam, Nielsen, Tapp, LNCS 1509, 61 (1998).

Take-home

Three 4-qubit QPUs reach about 0.9 on four digits, but the encoding did most of the work: communication needs a task where it must matter.

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Project results

My research notes on the DQML runs of 2026-09-25 to 09-28 and on MNIST-1D.

Thank you

Questions & discussion

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